

ARTICLE

Tonal languages without tone: downstep in Drubea and Numèè (Oceanic, New Caledonia)

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Abstract

In this article, I analyse the word-prosodic system of Drubea and Numèè, two of the rare tonal Oceanic languages. Building on Rivierre’s (1973) seminal work, I show that the word-prosodic system of these two languages can be analysed as involving only register features: an underlying downstep and a postlexical epenthetic upstep. Drubea and Numèè are thus tonal languages without tones *stricto sensu*. This new type of word-prosodic system has both theoretical and typological implications: (i) register features, defined as in Snider’s (1999) Register Tier Theory, need not be subordinate to or associated with tones, and may exist in the absence of tone, including in underlying representation; (ii) tonal systems come in two types: tone-based systems in which the tonal contrasts are defined paradigmatically, as in most tone languages, and register-based systems where tonal contrasts are defined syntagmatically, as in Drubea and Numèè.

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1. Introduction

This article is a contribution to our understanding of the cross-linguistic properties of register-affecting phenomena such as downstep and upstep, looking specifically at the rare and intriguing characteristics of downstep in Drubea and Numèè, two Oceanic languages of New Caledonia.

Downstep is a well-attested prosodic phenomenon affecting register – that is, the pitch range within which tonal contrasts are to be realised, defined by a ‘floor’ (lowest pitch within the range) and a ‘ceiling’ (highest pitch within the range). Specifically, downstep is a contrastive pitch drop that sets a new, lower register ceiling for the remainder of its domain, most of the time the utterance.¹ In most documented cases, downstep is caused by a low (L) tone and targets a following high (H) tone. It is customary to distinguish between ‘automatic’ downstep, caused by an overt L tone, for example, /HLH/ = [HL⁺H] = [l ɿ ɿ], and ‘non-automatic’ downstep when the L tone trigger is not heard, for example, when it is deleted or set afloat, as in /HLH/ = [H⁺H] = [l ɿ] (Stewart 1965). One of the main characteristics of languages with non-automatic downstep is the ‘terracing effect’ (Winston 1960; Connell 2011) created by multiple successive downsteps in an utterance, with each downstep bringing the pitch range (or register ceiling) down by one notch, for example, /HLHHHLHHH/ = [H⁺HHH⁺HHH] = [l ɿ ɿ ɿ ɿ ɿ ɿ]. Since automatic downstep is not at issue here, I will use ‘downstep’ to refer to the non-automatic kind only.

Downstep has been found to be triggered mostly by L tones, and to affect mostly H tones. Downstepped ⁺L and ⁺M tones are indeed very rare: I know of only 15 cases of the former in the literature,² and only eight languages are described as having a downstepped ⁺M tone to my knowledge.³ Cases of downstep not caused by L tones are rare, but attested. The downstepped L tone in Bamileke Dschang, for instance, has been analysed as being caused by a floating H (Hyman 1985b; Snider 1999, 2020). Cases where downstep is not caused by a floating tone are also attested, such as dissimilatory downstep between two H tones (/HH/ = [H⁺H]) in Shambala (Odden 1982) or Supyire (Carlson 1983). However, cases of downstep not triggered by a L tone or affecting other tones than H are still considered exceptional, and are consequently still understudied (cf. Leben 2018).

One additional point of interest is that downstep has until now been analysed only as a derived phenomenon emerging from tonal interaction, either in postlexical phonology or in phonetic implementation (see Connell 2011 for an overview). Until very recently, it had never been proposed that

¹ See Winston (1960); Stewart (1965, 1983, 1993); Clements (1979); Hyman (1979); Rialland (1997); Connell (2011); Hyman (2017); and Leben (2018); and the references therein.

² Bamileke Dschang (Tadadjeu 1974; Hyman 1985b; Bird & Stegen 1993, 1995), Bamileke Ghomala (Nissim 1981), Kikuyu (Clements & Ford 1977), Nawdm (Nicole 1980), Podoko (Anderson & Swackhamer 1981), Nandi (Creider 1982; Hyman 1984), Ghotuo (Elugbe 1986), Pāri (Andersen 1988), Edopi (Kim 1996), Saxwe (Beavon-Ham 2012), Igala (Adeniyi 2016), Yala (Adeniyi 2016), Sinyar (Boyeldieu 2019), Kugama (Litvinova 2023) and Paicî (Lionnet 2022b).

³ Peñoles Mixtec (Daly & Hyman 2007), Bagiro (Boyeldieu 2000), Ghotuo (Elugbe 1986), Gokana (Hyman 1985a), Klao (Lightfoot 1974), Yoruba (Pulleyblank 1986), Ngamambo (Hyman 1986) and Kugama (Litvinova 2023). I thank Larry Hyman (p.c.) for this information.

downstep could also be an underlying phonological primitive. I have shown elsewhere that in Paicî, another Oceanic language of New Caledonia, downstep is best viewed as a phonological object of its own, present in underlying representations (Lionnet 2022b). A similar claim has been made by Rochant (2023) about Baga Pukur, an Atlantic language of Guinea. In both cases, downstep is underlying in a system that also has underlying tones; for example, the three underlying prosodic primitives of Paicî are the two tones H and L and a downstep register feature. Underlying downstep is thus another attested, but non-canonical form of downstep.

The present article aims to contribute to a better understanding of non-canonical forms of downstep by offering a description and analysis of Drubea and Numèè, two very closely related Oceanic languages of New Caledonia whose word-prosodic system is typologically unique. Indeed, I show that the ‘tonal’ system of Drubea and Numèè, first expertly described by Rivierre (1973), can be analysed as consisting only of register features – specifically an underlying downstep and a postlexical epenthetic upstep – and no tone at all. The main underlying word-prosodic contrast is between downstepped register-bearing units (RBUs, defined as the mora, although the syllable also plays a role; see §4.2) and registerless RBUs, that is, a contrast between /⁺/ and ∅. Upstep is optionally inserted on registerless RBUs immediately before a downstep in order to increase the downward contrast marked by the downstep, which gives rise to alternations of higher and lower pitch heights and thus to the illusion of a tonal H–L contrast.

This system is particularly interesting from a cross-linguistic and typological perspective, as it is the only word-prosodic system known to date that rests entirely on register features, without any need for tones. This enriches our understanding of word-prosodic typology, in particular, the range and variety of ‘non-canonical’ tone systems in Hyman’s (2006) word-prosodic typology. This also calls into question the definition of ‘tone’ and what is required for a language to be considered ‘tonal’. Specifically, two types of tone systems must be recognised: in addition to the well-known TONE-based systems, in which tonal contrasts are defined paradigmatically (e.g., H is realised with higher pitch than L in the same context), REGISTER-based systems like that of Drubea and Numèè must also be recognised, in which tonal contrasts are defined syntagmatically (e.g., a downstepped unit is realised with lower pitch than the preceding unit). From a theoretical perspective, this register-only system is highly relevant for debates regarding the representation of tone and register, in particular, the existence of dedicated register features, proposed by Snider (1990, 1999, 2020), and the relation between register and tone.

I address all these questions in the article. I first give in §2 some relevant background information about Drubea and Numèè and the data that serve as the empirical basis of the main claim of the article. I then describe the word-prosodic system of the two languages in §3, and propose a register-based analysis in §4, couched in a modified version of Snider’s (1999, 2020) Register Tier Theory. This analysis is then shown to be superior to a purely tonal alternative in §5. The typological and theoretical implications of the recognition of register-based tonal languages such as Drubea and Numèè are then discussed in §6.

2. The data

2.1. Drubea and Numèè

Drubea [ŋaa ⁺nɖu^mbea] (Glottocode: dumb1241) and Numèè [ŋaa ⁺ɲumeɛ] (nucl1484) are the two southernmost languages of Grande Terre, New Caledonia’s main island.⁴ Together with Kwényĩ, spoken on the Isle of Pines, about 50 km southeast of Grande Terre, they constitute the Far South subgroup

⁴/ŋaa/ means ‘language’. The two language names are historically one and the same: Proto-Drubea–Numèè *⁺ɲumea changed into [⁺ɲumeɛ] in Numèè through vowel coalescence, and into [⁺nɖu^mbea] in Drubea through partial denasalisation of the nasal consonant followed by an oral vowel (see Rivierre 1973: 52–61, 98–104). Language names follow the orthography proposed by the Académie des Langues Kanak (2011, 2015, 2018). Note that Numèè speakers usually prefer to name their language variety after the place where it is spoken (e.g., /ŋaa xɛɛ/ ‘language of Goro’, /ŋaa ⁺ɲauɾu/ ‘language of Touaourou’, etc.). ‘Numèè’ is kept here as a convenient cover term, following Rivierre (1973); Wacalie (2013); Académie des Langues Kanak (2015). The ISO

(Haudricourt 1971) within the New Caledonian linkage of Southern Oceanic (Lynch *et al.* 2002), and count among the five tonal languages of New Caledonia, with Paicî and Cèmuhî, spoken in the central/northern region of Grande Terre.

Numèè and Kwényĩ are sometimes described as dialects of one language. I have excluded Kwényĩ from the present article because it differs markedly with respect to its prosodic system (Rivierre 1973: 154).⁵ Drubea and Numèè, on the other hand, share the same tone system (with only minor differences), which is the focus of this article.

Drubea is spoken by approximately 1,000 people.⁶ There are three dialectal variants: [ɲaa vũɲya] in Unya, on the east coast, as well as two dialects in Paita on the west coast – [ɲaa pweco] (lit. ‘language of the coast people’) on the coast, and [ɲaa ʔgakure] (lit. ‘language of the mountain people’) in the mountains.

Numèè is spoken by less than 1,000 speakers in three villages in the Far South of Grande Terre: Waho, Touaourou and Goro.⁷ Dialectal variation in Numèè is minimal (Académie des Langues Kanak 2015; Wacalie 2013).⁸

2.2. *Previous work and data sources*

Significant work on the Far South languages started with Rivierre’s (1973) description of the phonological systems of the Unya dialect of Drubea (/ɲaa vũɲya/), the variety of Numèè spoken in Goro (/ɲaa xeɾe/), and Kwényĩ.⁹

Additional work on Drubea includes Shintani and Païta’s grammar (1990b) and dictionary (1990a), based mostly on the mountain dialect of Paita, as well as a phonetic description of aspects of the consonant and vowel systems by Gordon & Maddieson (1999). Shintani also collected texts which he published in two separate volumes: Genet *et al.* (1992) and Shintani (2019). The latter is meant to serve as a language textbook, and was adapted to an online language tutorial in 2023 by the Académie des Langues Kanak (ALK), together with sound files of the original recordings (Académie des Langues Kanak 2023).¹⁰

Additional work on Numèè includes a morphosyntactic description by Wacalie (2013), an unpublished lexicon by Rivierre & Vandégou (n.d.), as well as an M.A. thesis by Rendina (2009), which I have not been able to consult. Texts recorded by A.-G. Haudricourt and J.-C. Rivierre in the 1960s

639-3 code for Drubea is ‘duf’; the one for Numèè (including Kwényĩ) is ‘kdk’. I use Glottocodes to clearly distinguish Numèè from Kwényĩ.

⁵Rivierre (1973: 154) states that ‘Numèè and Kwényĩ have 90% of their vocabulary in common, *modulo* regular vowel changes; but their sentence melody is astonishingly opposite. The first question that comes to mind is about the very existence of tone’ in Kwényĩ. (All translations from Rivierre 1973 are mine.) He then concludes that Kwényĩ is indeed tonal and sketches the main properties of the system: morphemes have the same underlying tonal specification as in Numèè, but their realisation is markedly different; it is highly context-dependent, and varies according to several criteria. His conclusion is that ‘the tonology [of Kwényĩ] appears as a structured and complex edifice, but threatened, undermined in many places, a possible illustration of a process whereby a two-tone system ceases to be functional’ (Rivierre 1973: 154). In her study of the phonology of Kwényĩ, Soon (2023: 137) concludes that ‘[a]lthough pitch contrast is observed in some minimal pairs, impressionistically tone has a low functional load and plays a marginal role in signifying lexical contrast. [...] This could indicate that tone is being lost as a contrastive feature in the language.’ The reader interested in the Kwényĩ prosodic system is referred to these two references for more detail.

⁶One thousand twenty-two speakers according to the 2019 census (INSEE-ISEE 2019).

⁷Numèè and Kwényĩ are not distinguished in census data. The 2019 census gives a global figure of 1,618 for both languages (INSEE-ISEE 2019), the majority of which are likely to be speakers of Kwényĩ.

⁸If one excludes Kwényĩ, or considers it to be a separate language.

⁹Previous work of more limited scope includes a 1,200-word list and a very brief morphosyntactic sketch of Drubea and Numèè in Leenhardt (1946), as well as short word lists collected by a dozen travellers who visited New Caledonia at various points in time (detailed in Rivierre 1973: 19–20). A.-G. Haudricourt also collected extensive lexical data and many texts in both languages in 1963 (Rivierre 1973: 20), some of which are available in the Pangloss collection (Haudricourt & Rivierre, n.d.).

¹⁰André-Georges Haudricourt and Jean-Claude Rivierre both collected texts in Drubea in the 1960s, but neither the recordings nor the transcriptions and translations of these texts are currently available, either in print or in accessible archival collections.

Table 1. *Adaptation of Rivierre's (1973) scale for pitch notation.*

	Rivierre	Retranscription
Highest	1	5
	1.5	4.5
	2	4
	2.5	3.5
	3	3
	3.5	2.5
	4	2
	4.5	1.5
Lowest	5	1

are available in the online Pangloss collection, where they are transcribed, glossed and translated into French (Haudricourt & Rivierre, *n.d.*).

I have not collected any data on either Drubea or Numèè. The present article is thus entirely based on secondary data – mainly Rivierre's (1973) phonological description and analysis of both languages; the Drubea grammar and Drubea-French dictionary published by Shintani & Païta (1990b,a); texts collected by Shintani and published in Shintani (2019) and Académie des Langues Kanak (2023); and Haudricourt & Rivierre's (*n.d.*) recordings of Numèè texts. For Shintani's Drubea recordings and transcriptions, two citations are given: Shintani's (2019) book, and the corresponding lesson number in the book and the online tutorial (Académie des Langues Kanak 2023), marked with an *L*. For example, 'Shintani 2019: 254, L35' refers to lesson 35, the text of which is found on page 254 of Shintani (2019), and the recording under the link labelled 'Leçon 35' in the online tutorial.¹¹ Speakers in Shintani's recordings were all women in their sixties and seventies at the time of recording in the 1990s: Augustine 'Titine' Betto (*née* Païta), Philomène 'Philo' Poarareu and Françoise Gaïa (*née* Païta).

Rivierre's (1973) description is not accompanied by recordings (if the recordings exist, they are not available in any known online or physical repository) which would allow one to check Rivierre's phonetic transcription of pitch heights. The analysis I present in this article thus partly rests on Rivierre's transcriptions. Rivierre is said by linguists who knew him to have had an excellent ear, which I was able to verify myself when working on Paicî, another New Caledonian language whose tone system he described and analysed (in Rivierre 1974). My findings concur with his almost perfectly, which gives me full confidence in his transcriptions. I was also able to find confirming evidence in Shintani's Drubea recordings and Rivierre's own recorded Numèè texts, both mentioned above.

Glosses for grammatical words follow Shintani & Païta's (1990b) grammatical analysis and terminology, including in examples borrowed from Rivierre (1973), whose glosses of function words lacks precision. Translations to English are my own.

Throughout the article, detailed information on the phonetic realisation of pitch contrasts of each example is systematically provided. This comprises the original narrow transcription of pitch heights for examples found in Rivierre (1973) and Shintani & Païta (1990b), and actual pitch tracings for examples taken from recorded texts (Shintani 2019; Académie des Langues Kanak 2018; Haudricourt & Rivierre, *n.d.*), for which only a broad, phonological transcription is given in the original source. All transcriptions drawn from original sources are converted to standard IPA notation (*modulo* a few simplifications; see §2.3). This includes the notation of pitch realisations: I have converted Shintani & Païta's schematic notation and Rivierre's numerical notation to the more standard numerical system,

¹¹The transcriptions in Shintani (2019) and in the online tutorial do not always correspond exactly to the accompanying recordings. The transcriptions proposed in this article are my own re-transcriptions of the recordings.

using 1 to represent the lowest pitch and 5 the highest. To be faithful to the original transcriptions, I retain Rivierre’s use of half steps (e.g., 1.5, 3.5), although these are not standard IPA. The correspondences between Rivierre’s original transcription and my retranscription are given in Table 1.

2.3. Phonological sketch

Drubea and Numèè (as well as Kwenyĩ) have virtually the same consonant inventory, presented in (1), where the phonological transcription used in this article (in both underlying and surface transcriptions) is accompanied by a phonetic transcription. The only difference is that Drubea systematically has /g^w/ where Numèè has /ŋ^w/. The plosive series is structured along a two-way contrast between voiceless and prenasalised voiced plosives, as in all the languages of Grande Terre. To simplify the transcriptions, prenasalised plosives will be systematically transcribed as plain voiced plosives in this article (e.g., /b/ = /^mb/), as is usual in New Caledonian linguistics.

(1) *Drubea and Numèè consonant inventory* (Rivierre 1973: 45–46; Shintani & Païta 1990b: 1)

Voiceless plosive	p	pw	t	t̚	c	k	kw
Prenasalised plosive	b [ᵐb]	bʷ [ᵐbʷ]	d [ᵐd]	d̚ [ᵐd̚]	j [ᵐj]	g [ᵐg]	gʷ [ᵐgʷ]†
Nasal	m	mw	n	ɳ	ɲ	ŋ	ŋw*
Fricative	v					x [χ]	
Flap				ɾ			
Glide					y [j]		w

*Only in Numèè.

†Only in Drubea.

The vowel inventories of the two languages are also very close, the only difference being the presence in Numèè of the three long front rounded vowels /üü øø õõ/, absent from Drubea, as shown in (2).

(2) *Drubea and Numèè vowel inventory* (Rivierre 1973: 82, 89; Shintani & Païta 1990b: 9)

	Short		Long		
Oral	i	u	ii	üü [y:]*	uu
	e	ɔ	ɛɛ		ɔɔ
	ɛ	o	ee	øø*	oo
	a		aa		
Nasal	ĩ	ũ	ĩĩ		ũũ
	ẽ	õ	ẽẽ	õõ*	õõ
	ã		ãã		

*Only in Numèè.

Rivierre (1973) only gives a partial account of syllable and word structure in Drubea and Numèè. He shows that coda consonants are not allowed: only open syllables are attested, with or without an onset consonant. Vowel length is said to be contrastive. Permissible syllable structures can thus be summarised by the formula (C)V(V), with VV representing a long vowel. According to Rivierre (1973), sequences of unlike vowels are heterosyllabic, never diphthongs, and there are no heterosyllabic sequences of identical vowels – that is, /kie/ [ki.e] ‘axe’ is disyllabic (not *[kiē]), whereas /kee/ [ke:] ‘husband’ is monosyllabic (and *[ke.e] is not attested). Shintani & Païta (1990b: 1) propose the same analysis for Drubea. I will follow this analysis here.

Most monomorphemic stems are mono- or disyllabic, as illustrated in (3), where σ and σ: stand for (C)V and (C)VV, respectively.¹²

¹²For the purposes of this article, the stem is defined as a free-standing monomorphemic lexical item.

- (3)
- | | | | |
|----|--------|-----------|------------------|
| a. | σ | /be/ | ‘death, die’ |
| | | /ĩ/ | ‘end, extremity’ |
| b. | σ: | /boo/ | ‘blind’ |
| | | /ʊʊ/ | ‘net’ |
| c. | σ.σ | /ku.ɾe/ | ‘forest, bush’ |
| | | /i.˥ya/ | ‘to fish’ |
| | | /kwi.e/ | ‘wind’ |
| | | /i.a/ | ‘declare war’ |
| d. | σ:..σ | /pwẽẽ.dĩ/ | ‘youngest son’ |
| | | /ũũ.ɾe/ | ‘moon’ |
| e. | σ.σ: | /wã.ɾee/ | ‘liana’ |
| | | /u.tii/ | ‘go home’ |
| | | /be.ii/ | ‘be jealous’ |
| f. | σ:..σ: | /vẽẽ.too/ | ‘fish sp.’ |

Stems of more than two syllables are also attested, although many appear to be morphologically complex – in particular, compounding (both in nouns and in verbs) is frequent. For lack of time and data, Rivierre does not provide a full analysis on the internal structure of words of more than two syllables. He mostly ignores these words in his analysis of the tone system, which I will also do, for the same reasons (cf. §3.8).

3. The word-prosodic system of Drubea and Numèè

As shown by Rivierre (1973: 124–153), the word-prosodic systems of Drubea and Numèè are virtually identical. The only significant difference lies in utterance-final prosodic phenomena, discussed in §3.4. Rivierre describes the two systems as one, illustrating his generalisations and analyses with examples taken from both. I do the same here, although I give more Drubea than Numèè examples, mainly because of the relative abundance of available Drubea data. The source language is always explicitly given in all examples.

In this section, I give a description of the surface prosodic patterns attested in both languages, using analytical categories (in particular, ‘downstep’ and ‘upstep’) that are fully developed in the analysis I propose in §4. I first describe the tonal behaviour of monosyllabic stems (henceforth, ‘monosyllables’; §§3.1–3.5), then that of disyllables (§3.6), before discussing the special case of CV⁺V syllables (§3.7). The issue of stems of more than two syllables is briefly addressed in §3.8, and morphophonological tonal effects in §3.9.

3.1. Downstepped vs. registerless monosyllabic stems

Monosyllables mostly fall into two prosodic types in Drubea and Numèè: those that are systematically realised lower than the preceding morpheme, and those that are not.¹³ The latter tend to be realised at the same pitch as the preceding morpheme, all else being equal, but are also very frequently realised with higher pitch, as we will see in §3.2. I will, for now, call these two types SAME-PITCH and LOWER-PITCH syllables, respectively, and will transcribe the latter with a preceding downstep mark, a transcription which reflects the analysis I propose in §4. Note that I use the term ‘syllable’ in this section for ease of exposition. I will show in §§3.7 and 4.2 that the tone- (or rather register-) bearing unit is actually the mora (although the syllable also has a role to play).

Numerous minimal pairs are attested in both languages, as illustrated with Drubea examples in (4) (Rivierre 1973: 123–124; Shintani & Païta 1990b: 17).

¹³There is in fact a third, minor type, as we will see in §3.7.

- (4)
- | | | | |
|----------|---------------------|--------|---|
| a. /ĩ/ | ‘extremity, tip’ | /˚ĩ/ | ‘piece, bit’ |
| b. /ɪɪ/ | ‘indigenous bamboo’ | /˚ɪɪ/ | ‘ <i>Elaeocarpus angustifolius</i> , tree sp.’ |
| c. /be/ | ‘death; to die’ | /˚be/ | ‘ <i>Melaleuca quiquenervia</i> , niaouli’ |
| d. /doo/ | ‘bag, envelope’ | /˚doo/ | ‘ <i>Cordyline</i> spp., plant sp.’ |
| e. /cu/ | ‘to wipe’ | /˚cu/ | ‘to knock down with a pole’ |
| f. /ɲi/ | ‘coconut tree’ | /˚ɲi/ | ‘breast’ |
| g. /kee/ | ‘husband’ | /˚kee/ | ‘ <i>Broussonetia papyrifera</i> , mulberry tree’ |

The contrast between same-pitch /be/ ‘die’ and lower-pitch /˚be/ ‘niaouli tree’ is illustrated in (5) and (6) (the morpheme under consideration is underlined in the examples). As can be seen in Figures 1 and 2, /be/ ‘die’ is realised at the same pitch as the preceding syllable, while /˚be/ ‘niaouli tree’ is realised at a lower pitch.¹⁴

- (5) /... ˚mwa be to ˚ɲe co + ˚%/
- [... ˚mwã be to ˚ɲe ˚%co]
- PFV die LOC in water
- ‘[his son] died in the water’ (Drubea; Shintani 2019: 178, L24)
- (6) /dɪɪ ˚be/
- [˚dɪɪ ˚be]
- small niaouli
- ‘small niaouli tree’ (Drubea; Shintani 2019: 353, L49)

This same-pitch vs. lower-pitch realisation is seen irrespective of the prosodic type of the preceding syllable. Examples (7) and (8) illustrate the descriptive marker /te/ following the same-pitch noun /aboɲu/ ‘person’ in (7), and following the lower-pitch perfective marker /˚mwa/ in (8).¹⁵ Figures 3 and 4 show that /te/ is realised at the same pitch as the immediately preceding syllable in both cases, irrespective of whether it is a same-pitch or lower-pitch syllable. Note that the speaker repeats the marker /te/ in (7), and both instances are realised at the same pitch as the last two syllables of the noun /aboɲu/. The pitch rise seen in [a˚boɲu] need not concern us here, and will be dealt with in §3.2. The

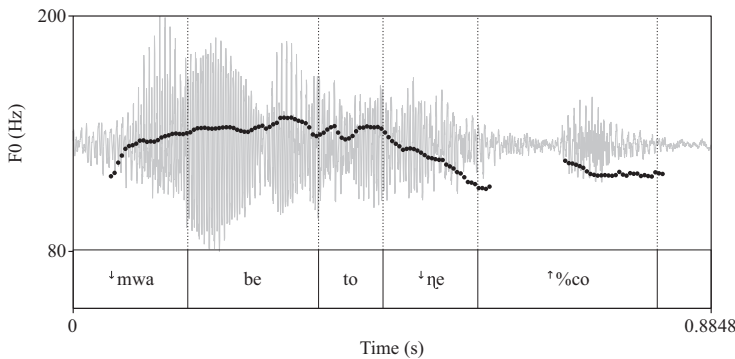


Figure 1. /... ˚mwa be to ˚ɲe co + ˚%/ ‘[his son] died in the water’ (Drubea; Shintani 2019: 178, L24).

¹⁴The upstep seen in [˚dɪɪ] in (6) is discussed in §3.2. The utterance-final boundary upstep ˚% in (5) is explained in §3.3. Notice also the pitch-lowering effect of utterance-initial nasal stops, in both [˚mwa] in Figure 1 and the prenasalised portion of [˚dɪɪ] in Figure 2. This lowering effect is frequent in Drubea and Numèè, and will be seen recurrently on pitch tracings in the remainder of the article.

¹⁵Shintani & Paita (1990b: 25) define the ‘descriptive’ tense-aspect category as ‘expressing an action or notion that is specific and conceived of as taking place in the present’.

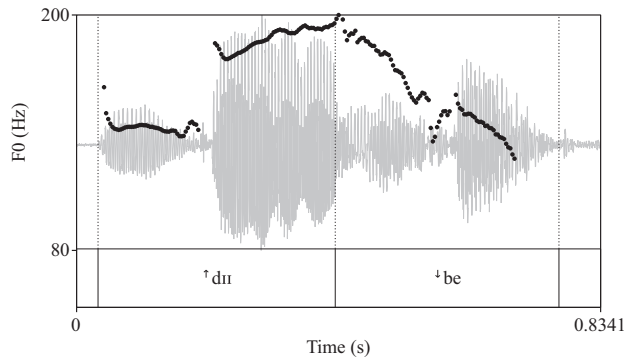


Figure 2. /dui ʔbe/ 'small niaouli' (Drubea; Shintani 2019: 353, L49).

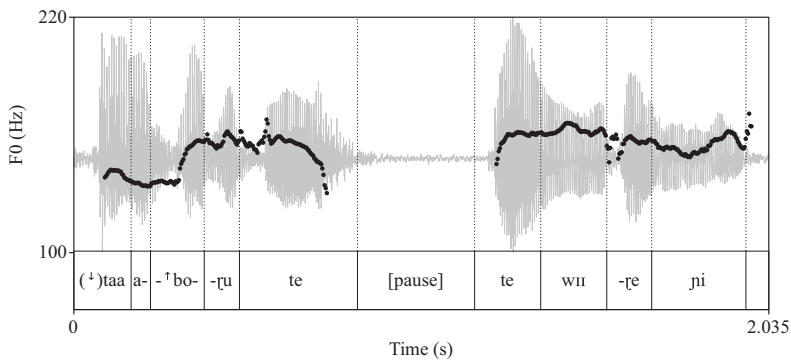


Figure 3. /ʔtaa aboɾu te || te wu-ɽe ji/ 'someone is squeezing coconut pulp' (Drubea; Shintani 2019: 172, L23).

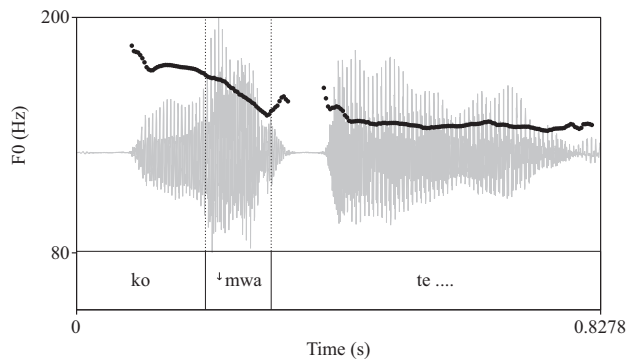


Figure 4. /ko ʔmwa te .../ 'I [work my field]' (Drubea; Shintani 2019: 234, L32).

important point is that there is no pitch drop in the transition from the last syllable of /aboɾu/ to the following same-pitch syllable /te/.¹⁶

- (7) /⁺taa aboɾu te || te wɪɪ-ɾe ni/
 [(⁺)taa a⁺boɾu te || te wɪɪ-ɾe ni]
 one person DESCR DESCR squeeze-ACT coconut
 ‘someone is... is squeezing coconut pulp’ (Drubea; Shintani 2019: 172, L23)

- (8) /ko ⁺mwa te ɲa-ɾe... /
 [ko ⁺mwã te ɲa-ɾe...]
 ISG.SBJ PFV DESCR do-ACT
 ‘I work [my field]’ (Drubea; Shintani 2019: 234, L32 AT)

Examples (9) and (10) illustrate the lower-pitch word /⁺tã/ ‘fire(wood)’ after same-pitch /kaa/ ‘smoke’ (9) and lower-pitch /⁺vi/ ‘take’ (10). As can be seen in Figures 5 and 6, /⁺tã/ is realised lower than the preceding syllable in both cases.¹⁷

- (9) /... kaa ⁺tã/
 [... ⁺kaa ⁺tã]
 smoke fire
 ‘...(fire) smoke’ (Drubea; Shintani 2019: 173, L23)
- (10) /kã⁺ã ⁺vi ⁺tã/
 [kã⁺ã ⁺vi ⁺tã]
 PROS take firewood
 ‘[then we] take firewood’ (Drubea; Shintani 2019: 172, L23)

A sequence of same-pitch syllables is realised, unsurprisingly, with the same pitch throughout, as in the sequence [...boɾu te || te wɪɪ-ɾe ni] in (7) and Figure 3. A sequence of lower-pitch syllables, on the other hand, is realised as a series of pitch drops (Rivierre 1973: 145; Shintani & Païta 1990b: 19). This is already illustrated in (10) and Figure 6. Two additional examples are given in (11) and (12).¹⁸

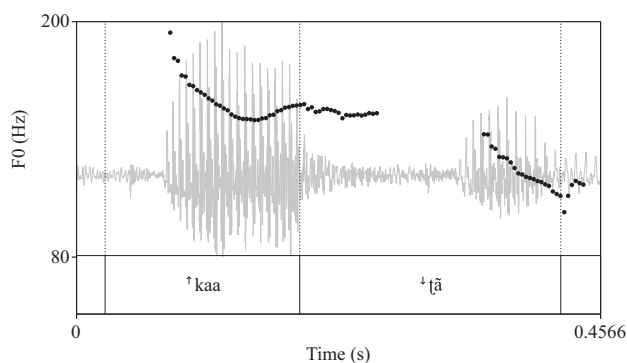


Figure 5. /kaa ⁺tã/ ‘(fire) smoke’ (Drubea; Shintani 2019: 173, L23).

¹⁶The utterance-initial realisation of lower-pitch syllables, such as /⁺taa/ in (7), will be discussed in §§3.5 and 4.5.

¹⁷The slightly falling melody on [kaa] in (9) and Figure 5 is unexplained. It does not appear to be the realisation of any category or meaning, and is barely audible in the recording.

¹⁸In examples taken from Rivierre (1973) and Shintani & Païta (1990b), the original transcription is transposed into numerical notation on the third line (see Table 1), and reinterpreted within the analysis proposed in this article on the first two lines.

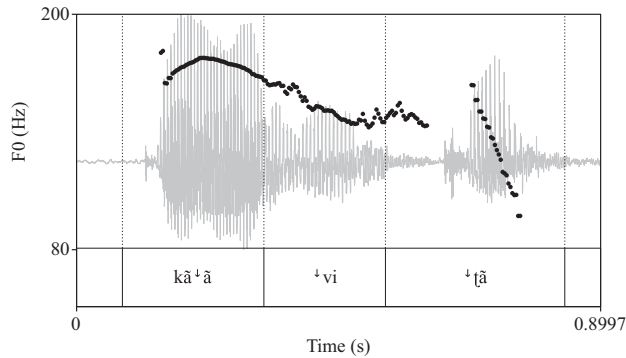


Figure 6. /kã˥˥ vi ˥[ã/ ‘[then we] take firewood’ (Drubea; Shintani 2019: 172, L23).

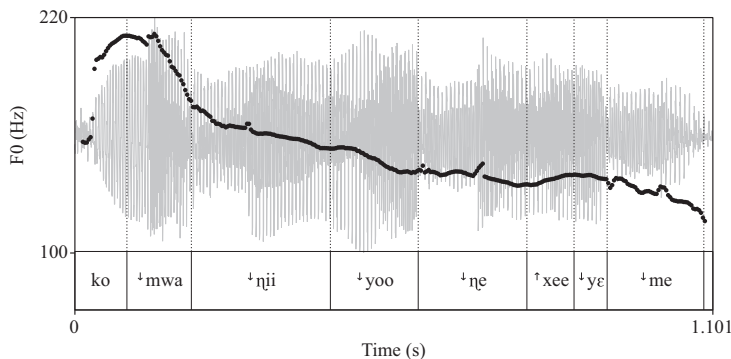


Figure 7. /ko ˥mwa ˥ɲii ˥yoo ˥ne-xee-˥ye ˥me.../ ‘I said just now that...’ (Drubea; Shintani 2019: 306, L43 AT).

- (11) /˥ɲi ˥mwa ˥ɲii ˥me/
 [(˥)ɲi ˥mwa ˥ɲii ˥me]
 [ɲi⁴ mwa³ ɲii² me¹]
 3PL.SBJ PFV say that
 ‘They said that...’ (Drubea; Shintani & Païta 1990b: 19)

Example (12) shows a succession of four lower-pitch syllables: /... ˥mwa ˥ɲii ˥yoo ˥ne.../, followed by the same-pitch word /xee/, itself followed by two lower-pitch syllables /... ˥ye ˥me/.

- (12) /ko ˥mwa ˥ɲii ˥yo-V ˥ne-xee-˥ye ˥me .../
 [ko ˥mwa ˥ɲii ˥yoo ˥ne-˥xee-˥ye ˥me ...]
 ISG.SBJ PFV say to-2SG now that
 ‘I said just now that...’ (Drubea; Shintani 2019: 306, L43 AT)

Figure 7 shows that the first four consecutive lower-pitch syllables [˥mwa ˥ɲii ˥yoo ˥ne] in example (12) are realised with four consecutive drops in pitch. The successive pitch drops do not all have the same magnitude: the first drop [ko ˥mwa] is sharp, while the following ones are less and less perceptible, because the intervals are successively smaller as the speaker reaches the lower end of her pitch range. The slight pitch rise seen on [˥xee] will be explained in §3.2.

The behaviour of these two types of syllables suggests a contrast in register rather than in tone height: the lower-pitch syllables are realised in a lower register than the immediately preceding one.¹⁹

¹⁹ ‘Register’ is taken here as referring to pitch range, characterised by a floor and ceiling; it does not refer to phonation type, as is frequent in the literature on Southeast Asian languages, or any other attested use of the term.

This lower register is maintained for the remainder of the utterance, unless another lower-pitch syllable imposes a new register lowering. Furthermore, lower-pitch syllables are not associated with any particular pitch range: they are often realised toward the higher end of a speaker's pitch range at the beginning of an utterance, especially utterances including many lower-pitch syllables, that is, many pitch drops. Rivierre (1973: 153), who analyses lower-pitch syllables as low-toned and same-pitch ones as high-toned, as we will see in §5, says this explicitly: '[t]his contrastive nature [i.e., syntagmatic register contrast with the preceding syllable] of the low tone [= lower-pitch syllable] is more essential to its definition than its pitch height'. In other words, lower-pitch syllables behave like downstepped syllables. Same-pitched syllables, on the other hand, are inert from the point of view of register: they do not impose any register change, and their realisation is subject to contextual variation, as we will see in the next section. This suggests that they carry no underlying indication of pitch realisation. 'Lower-pitch' syllables will henceforth be referred to as 'downstepped', and 'same-pitch' ones as 'registerless', in accordance with the phonological transcription used so far, and the analysis fully developed in §4.

With only registerless and downstepped syllables, one should expect the pitch of an utterance to only ever go down – abstracting away from possible effects of intonation. This is mostly the case – and is, indeed, one salient characteristic of both Drubea and Numèè, which makes them noticeably different from the very closely related Kwényi (which, as noted by Rivierre 1973: 154, is characterised by an overall rising melody profile). One major exception to this principle is the raising frequently undergone by registerless syllables in two contexts: before a downstepped syllable, and utterance-finally (in Drubea), as discussed in the following two sections.

3.2. Pre-downstep raising of registerless syllables

As noted by Rivierre (1973: 132), before a downstepped syllable, speakers oscillate between two realisations of what I analyse as registerless syllables: either at the same pitch as the preceding syllable, as with the descriptive marker /te/ in (13a), or higher than the preceding syllable as in (13b), the latter being more frequent according to Rivierre (1973), which is confirmed by Shintani's recorded texts.

- (13)
- | | | | |
|----|--|------------------------|--|
| | /ko | <u>te</u> | *be <u>ɽ</u> u- <u>ɽ</u> e/ |
| a. | [ko | <u>te</u> | *be <u>ɽ</u> u- <u>ɽ</u> e] |
| | [ko ⁴ | <u>te</u> ⁴ | be ³ ɽu ^{2.5} -ɽe ^{2.5}] |
| b. | [ko | [↑] <u>te</u> | *be <u>ɽ</u> u- <u>ɽ</u> e] |
| | [ko ⁴ | <u>te</u> ⁵ | be ^{2.5} ɽu ² -ɽe ²] |
| | ISG.SBJ | DESCR | swim-ACT |
| | 'I swim.' (Drubea; Rivierre 1973: 132) | | |

This pre-downstep raising seen in (13b), which I transcribe and analyse as upstep (see §4.4), very frequently affects registerless syllables. As clearly described by Rivierre (1973: 153), this raising 'reinforces and highlights the contrast marked by each low tone [= downstep] with the preceding syllable. [...] A high-toned [= registerless] syllable is itself best defined as a syllable that does not mark a contrast of this type than as a high-pitched syllable' (Rivierre 1973: 153). Note that pre-downstep raising rarely affects downstepped syllables, and when it does, it is realised differently, as we will see below (see example (18) and the adjacent text). This is one of the key differences between the two types of syllables.

Rivierre (1973: 133) further notes that 'the raised realisation of the H tone [= registerless syllable] before a L tone [= downstep] is *common* but not obligatory; understandably, it is largely used in H–L–H–L [= registerless–downstepped–registerless–downstepped] sequences' (emphasis in the original). Without this raising of registerless syllables, 'each new L tone [= downstep] brings the speaker down toward lower and lower registers, making the register drop characteristic of the downstep more and more difficult to produce and perceive. The [raising of] H tones [=registerless syllables] is thus used in this context to maintain the register of the utterance within a range conducive to an economic

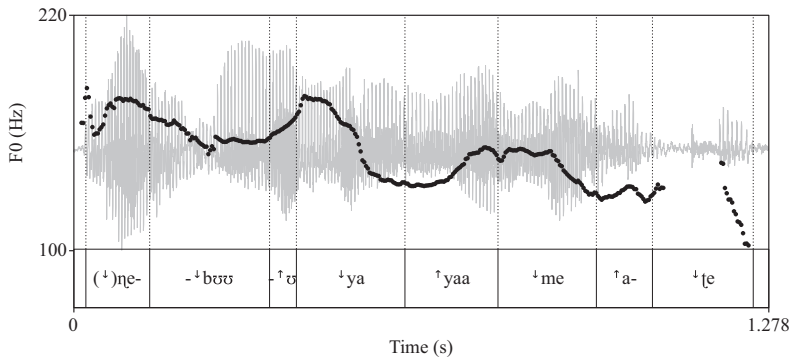


Figure 8. /[↑]ne-[↓]buu-V [↑]ya yaa [↑]me a-[↓]te/ ‘Your face does not look good’ (Drubea; Shintani 2019: 130, L17).

and perceptible realisation of L tones [=downsteps]’ (Rivierre 1973: 133). Rivierre (1973: 134) finally notes that ‘the raising of H-toned [= registerless] syllables rarely compensates for the lowering of the register caused by the preceding L tone [= downstep], which explains the overall descending trend in a downstepped–registerless–downstepped... sequence’. This is illustrated in (14) and (15). As seen in Figure 8, the successive valleys and peaks marked by the sequence of downsteps and upsteps in (15) are characterised by lower and lower pitch, and the overall pitch trajectory throughout the utterance follows a downward course.

- (14) /ko [↑]mwa cu[↑]kwa ð [↑]keʔee-ʔe/
 [↑ko [↑]mwa ↑cu[↑]kwa ↑ð [↑]keʔee-ʔe]
 [ko⁵ mwa⁴ cu^{4.5}kwa³ ʔ^{3.5} ke²ʔee²-ʔe]
 ISG.SBJ PFV finish LOC eat-ACT
 ‘I am done eating.’ (Drubea; Rivierre 1973: 133)

- (15) /[↑]ne-[↓]buu-V [↑]ya yaa [↑]me a-[↓]te/
 [([↑])ne-[↓]buu-[↑]u [↑]ya [↑]yaa [↑]me [↑]a-[↓]te]
 COLL-face-2SG.POSS then NEG EPIST STAT-good
 ‘Your face does not look good.’ (Drubea; Shintani 2019: 130, L17)

Pre-downstep raising is not always limited to the syllable that immediately precedes the downstep; it sometimes extends through a sequence of registerless syllables. This pitch raising scoping over several successive registerless syllables may be abrupt, as in (16) and Figure 9, where one can see that the raising effect caused by the downstep on the second mora of the determiner /ma[↑]a/ does not affect only the preceding registerless mora, but extends to the entire sequence of preceding registerless syllables, with the same raised pitch throughout.²⁰

- (16) /[↑]tā [↑]mwa ne-ʔe ma[↑]a vuu [↑]ɲia/
 [([↑])tā [↑]mwa [↑]ne-ʔe ma[↑]a vuu [↑]ɲia]
 fire PFV burn-ACT DET.PL stone DIST
 ‘The fire burns these stones.’ (Drubea; Shintani 2019: 172, L23)

Alternatively, the raising effect may be gradual, in which case, the pitch starts rising on the first registerless syllable of the affected sequence, and reaches its peak on the registerless syllable immediately before the downstep. This is illustrated in (17) and Figure 10, where the rise starts on the registerless syllable [pwe] immediately after the downstepped syllable [[↑]ɲo], and rises gradually until it

²⁰Syllable-internal downsteps such as in /ma[↑]a/ will be discussed in §3.7.

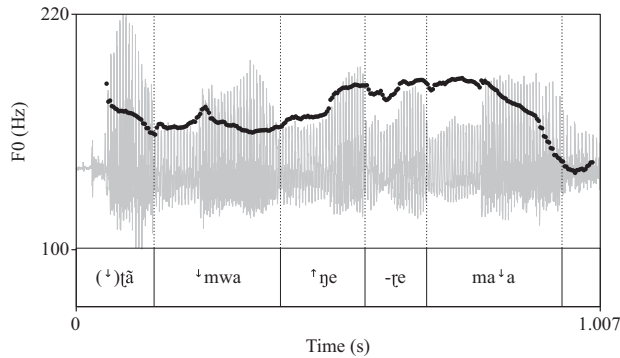


Figure 9. /[˩]tã [˩]mwa ɲe-ɲe ma [˩]a.../ ‘The fire burns these [stones]’ (Drubea; Shintani 2019: 172, L23).

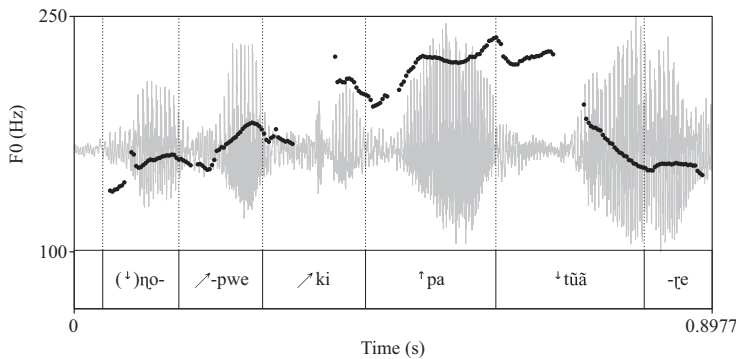


Figure 10. /[˩]ɲopwe ki pa [˩]tũã-ɲe.../ ‘And you see [that it is good]...’ (Drubea; Shintani 2019: 41, L04).

reaches its peak on the assertive marker [pa], immediately before the downstepped syllable [[˩]tũã] ‘see’. This gradual rise can be seen as a form of interpolation, that is, a gradual transition from the lower pitch of a downstepped syllable to the higher pitch of the next raised pre-downstep syllable. I transcribe this interpolation phenomenon with a northeast-pointing arrow [↗].

- (17) /[˩]ɲopwe ki pa [˩]tũã-ɲe pa a-[˩]ɲe/
 [([˩])ɲo↗pwe ↗ki ˩pa [˩]tũã-ɲe pa a-[˩]ɲe]
 and 2SG.SBJ ASSERT see-ACT ASSERT STAT-good
 ‘And you think it is a good thing?’ (lit. ‘and you see [that] it is good?’; Drubea; Shintani 2019: 41, L04)

Finally, as mentioned above, pre-downstep raising is also sometimes seen to affect downstepped syllables. In such cases, the downstep is never undone by the raising effect. Instead, the rise in pitch is heard on the end of the syllable, after the register lowering triggered by the downstep, in what can be described as a rising contour. This is attested even on monomoraic CV syllables, as seen in (18). ([V[˩]V]) represents a monomoraic vowel affected by upstep on its second half.)

- (18) /[˩]ɲɛ [˩]mwa [˩]vɛ/
 [([˩])ɲɛ [˩]mwa[˩] [˩]vɛ]
 [ɲɛ⁴ mwɑ^{3.5-4} vɛ³]
 3PL.SBJ PFV go
 ‘They go.’ (Numèè; Rivierre 1973: 147)

The realisation of registerless syllables is also sensitive to utterance-final phenomena, as seen in the next two sections: final raising in Drubea (§3.3), and final downstepping of light syllables in Numèè (§3.4).

3.3. Utterance-final raising in Drubea

A tendency toward raising of registerless syllables is also seen in utterance-final position in Drubea (Numèè differs from Drubea on this point, as discussed in the next section). This is particularly marked when the final syllables are preceded by one or more pitch drops caused by downstepped syllables. This effect is not explicitly described by either Rivierre (1973) or Shintani & Païta (1990b), but it can be seen in some of their transcriptions. Optional final raising is illustrated with the two examples below, from Rivierre (1973). In both cases, the utterance ends with a sequence of two registerless syllables. In (19), there is no final raising, while in (20), the final registerless syllable is realised at a higher pitch than the immediately preceding one. I transcribe this final pitch rise with an upstep symbol followed by a percent sign ($\uparrow\%$) at the end of the utterance in phonological transcription, and immediately before the final syllable in surface transcription. This transcription corresponds to the analysis of this rise as a final boundary upstep, which I propose in §4.8.

- (19) / $\uparrow$ taa dɪ bee/
 [($\uparrow$)taa dɪ bee]
 [taa⁴ dɪ⁴ bee⁴]
 one small fish
 ‘one small fish’ (Drubea; Rivierre 1973: 126)
- (20) / $\uparrow$ taa bee pwi + $\uparrow\%$ /
 [($\uparrow$)taa bee $\uparrow\%$ pwi]
 [taa⁴ bee⁴ pwi⁵]
 one fish cooked
 ‘one cooked fish’ (Drubea; Rivierre 1973: 128)

In the Unya dialect described by Rivierre (1973), this effect seems to be rare – at least the reader is led to infer so, as utterance-final registerless syllables are realised with the same pitch as the preceding syllable in all but two examples, of which (20) is one. Final raising is much more frequent in the Païta dialect. This is not explicitly said in Shintani & Païta’s (1990b: 17–22) description, but it can be seen in the few phonetic transcriptions they give. A cursory survey of the 50 texts recorded by Shintani and available in the online Drubea tutorial (Académie des Langues Kanak 2023) suggests that the vast majority of utterance-final registerless syllables undergo this final raising in the Païta dialect. The example in (21) is taken from one of these texts. As can be seen in Figure 11, the final two syllables [pwe wẽ] are clearly realised with a higher pitch than the preceding registerless prefix *a-*, which is pronounced at the same lowered pitch as the downstepped final syllable of the verb [*u⁺i*] immediately before it (see also (7)).²¹

- (21) /ke⁺e kã⁺ã u⁺i a-pwe wẽ + $\uparrow\%$ /
 [ke⁺e $\uparrow$ kã⁺ã u⁺i a- $\nearrow$ pwe $\uparrow\%$ wẽ]
 IPL.SBJ PROS do STAT-all LOC
 ‘We will do everything that way.’ (Drubea; Shintani & Païta 1990b: 172, L23)

3.4. Utterance-final downstepping in Numèè

In Numèè – at least in the Goro dialect described by Rivierre (1973) – utterance-final light CV syllables are systematically downstepped if they are preceded by a registerless syllable; that is, CV $\rightarrow$ $\uparrow$ CV / $\sigma_ \#$

²¹The downstep in the CV⁺V monosyllables /ke⁺e/ and /kã⁺ã/ will be discussed in §3.7.

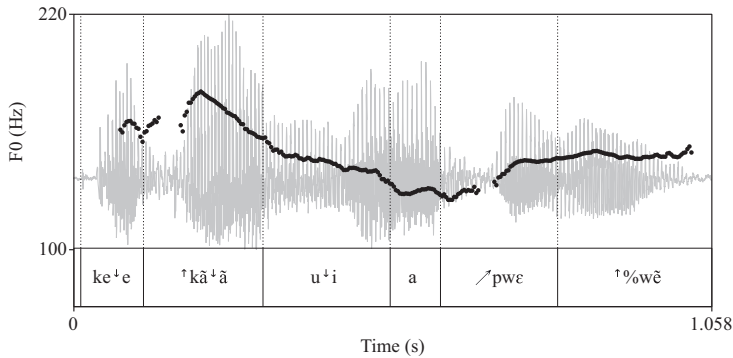


Figure 11. /ke⁺e kã⁺ã u⁺i-a -wε wẽ + ⁺%/ ‘We will do everything that way’ (Drubea; Shintani 2019: 172, L23).

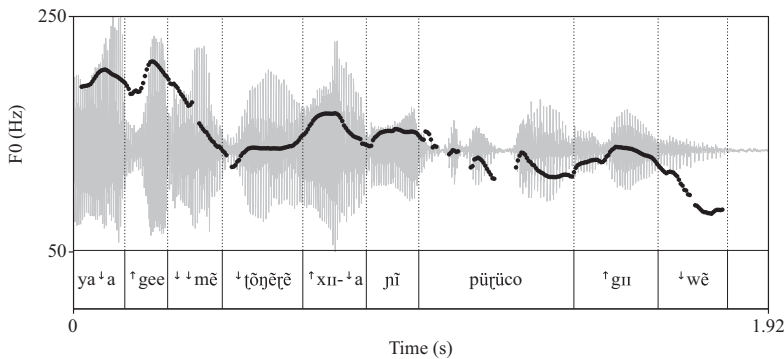


Figure 12. /... ya⁺a ge⁺e ⁺mẽ ⁺tõñẽrẽ xii-⁺a nĩ pũrũco gu wẽ + ⁺%/ ‘[this child], we don’t know where he’s coming from!’ (Numèè; Ati & Rivierre 1966: S54, 4:03).

(where σ stands for a registerless syllable). That this is indeed downstep is revealed by the fact that the preceding registerless syllable often undergoes pre-downstep raising, as is the case in (22) and in (23) and Figure 12, where the downstepping of final /ŋa/ → [⁺ŋa] and /wẽ/ → [⁺wẽ] triggers raising of the preceding /a/ → [⁺a] and /gu/ → [⁺gu], respectively. I analyse this final downstepping process as the realisation of a final boundary downstep, transcribed /⁺%/ (cf. §4.8).²²

- (22) /dɛŋu a ŋa + ⁺%/
 [dɛŋu ⁺a ⁺%ŋa]
 [dɛ⁴ŋu⁴ a^{4.5} ŋa³]
 jaw REL up
 ‘upper jaw’ (Numèè; Rivierre 1973: 132)

- (23) /... ya⁺a ge⁺e ⁺mẽ ⁺tõñẽrẽ xii-⁺a nĩ pũrũco gu wẽ + ⁺%/
 [... ya⁺a ⁺gee ⁺mẽ ⁺tõñẽrẽ ⁺xii-⁺a nĩ pũrũco ⁺gu ⁺wẽ]
 NEG IPL.INCL.SBJ NEG know place-DIST 3SG.SBJ come.out from LOC
 ‘[this child], we don’t know where he’s coming from!’ (Numèè; Ati & Rivierre 1966: S54, 4:03)

Additional evidence in favour of viewing final downstepping as the assignment of a boundary downstep is that it affects underlyingly downstepped syllables as well, as shown by the fact that the contrast between the two types of syllables is maintained in utterance-final position. An utterance-final

²²The non-initial downstep in /ya⁺a/ and /ge⁺e/, and the double downstep on [⁺mẽ] in (23), will be explained in §3.7.

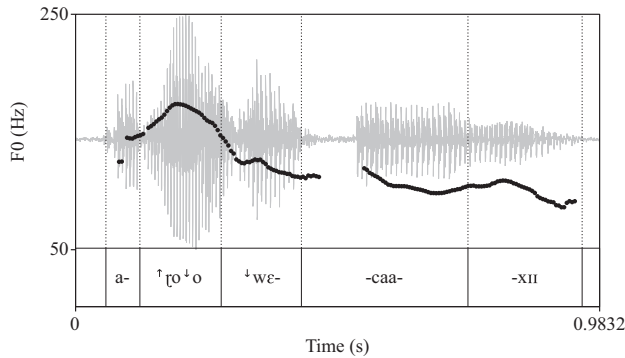


Figure 13. /a-[↑]ɾoo [↓]wɛcaaxɪɪ/ ‘... down near the channel’ (Numèè; Ati & Rivierre 1966: S31, 1:58).

registerless light syllable is indeed not realised as low as a downstepped syllable in the same position, as noted by Rivierre (1973: 127) and illustrated in (24) and (25). This shows that downstepping is the realisation of an inserted extra downstep, not just the result of a change of the final syllable from registerless to downstepped. When realised on an already downstepped syllable, the boundary downstep results in a double downstep, which explains the lower realisation of underlying downstepped /[↑]nĩ/ → [[↑]nĩ] in (25) compared to registerless /nĩ/ → [nĩ] in (24).

- (24) /jaa nĩ + [↑]%/
 [↑jaa [↑]%nĩ]
 [jaa⁵ nĩ⁴]
 juice coconut
 ‘coconut juice’ (Numèè; Rivierre 1973: 127)

- (25) /jaa [↑]nĩ + [↑]%/
 [↑jaa [↑]%[↑]%nĩ]
 [jaa⁵ nĩ^{3.5}]
 juice breast
 ‘breast milk’ (Numèè; Rivierre 1973: 127)

Final downstepping affects only light syllables: CVV syllables remain registerless utterance-finally, as in (26) (cf. (22)) and in (27) and Figure 13.²³

- (26) /dɛŋʊ a mɪj + [↑]%/
 [dɛŋʊ a mɪj]
 [dɛ⁴ŋʊ⁴ a⁴ mɪj⁴]
 jaw REL low
 ‘lower jaw’ (Numèè; Rivierre 1973: 132)

- (27) /a-[↑]ɾoo [↓]wɛcaaxɪɪ + [↑]%/
 [a-[↑]ɾo[↑]o [↓]wɛcaaxɪɪ]
 LOC-down near.the.channel
 ‘... down near the channel’ (Numèè; Ati & Rivierre 1966: S31, 1:58)

²³Note that the realisation of the downstep (and preceding upstep) in (27) and Figure 13 is delayed by one mora in /a-[↑]ɾoo/ → [a-[↑]ɾo[↑]o] (instead of the expected [[↑]a-[↑]ɾoo]). It is not clear why this is the case. This might be an example of pitch target delay, or carryover effect, very frequent in tone languages (see Hyman & Schuh 1974; Han & Kim 1974; Xu 1997; *inter alia*).

Finally, downstepping does not apply after a downstepped syllable, as shown in (28) and in (29) and Figure 14.

- (28) /⁺tēē-⁺ē nō bε⁺tīī ku + ⁺%/
 [(⁺)tēē-⁺ē nō ⁺bε⁺tīī ku]
 [tēē⁵-ē⁴ nō⁴ bε⁵⁺tīī³ ku³]
 girl-PROX grill three yam
 ‘This girl is grilling three yams.’ (Numèè; Rivierre 1973: 135)

- (29) /... nī yoo a-⁺paa kwē + ⁺%/
 [... nī /yoo ⁺a-⁺paa kwē]
 3SG.SBJ berth LOC-up sand
 ‘... he berths on the sand’ (Numèè; Ati & Rivierre 1966: S64, 4:57)

The latter examples suggest that the final boundary downstep /⁺%/ is realised only on final light syllables following a registerless syllable, and left unrealised in all other contexts. From now on, the boundary downstep will be included in underlying form only when it is realised.

3.5. Utterance-initial downstep

Downstep is not realised utterance-initially, where downstepped and registerless syllables are not phonetically distinct (unless they are followed by a downstepped syllable, as we will see below). This can be seen by comparing (30), which contains only registerless syllables, with (19), repeated here as (31), in which the initial syllable is downstepped. The two utterances start at the same pitch height. Non-realisation of the underlying downstep in (31) is indicated by parentheses.

- (30) /ko te tɪ-tɛ kuɾe/
 [ko te tɪ-tɛ kuɾe]
 [ko⁴ te⁴ tɪ⁴-tɛ⁴ kuɾe⁴]
 1SG.SBJ DESCR look.at-ACT bush
 ‘I look at the bush.’ (Drubea; Rivierre 1973: 127)
- (31) /⁺taa dɪ bee/
 [(⁺)taa dɪ bee]
 [taa⁴ dɪ⁴ bee⁴]
 one small fish
 ‘one small fish’ (Drubea; Rivierre 1973: 126; repeated from (19))

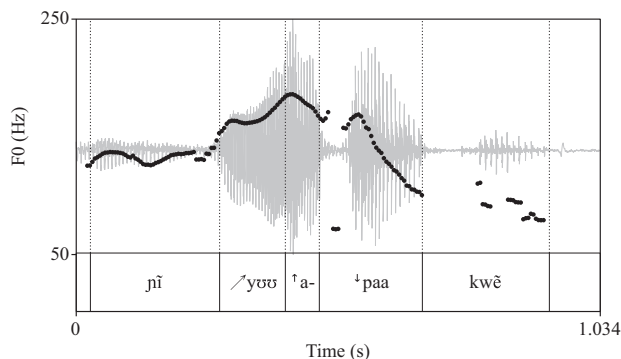


Figure 14. /... nī yoo a-⁺paa kwē/ ‘... he berths on the sand’ (Numèè; Ati & Rivierre 1966: S64, 4:57).

This is easily explained if downstep is viewed as the realisation of an instruction to contrast downward with the immediately preceding syllable. In the absence of any preceding syllable, there is nothing to contrast with, and the utterance starts at what can be considered to be the pitch baseline. According to Rivierre and Shintani's transcriptions as well as the selection of recordings from Shintani's (2019) text collection that I was able to process, the baseline in Drubea and Numèè can be schematically represented as mid-high, or 4 out of 5 on the pitch-height scale. This fairly high pitch level corresponds to the default pitch at the beginning of an unmarked utterance. The fact that the utterance starts by default at a rather high pitch is most likely meant to allow for a pitch decline in the course of the utterance, either because of the presence of following downstepped syllables (and the overall downward realisation of most utterances, noted by Rivierre 1973: 153), or simply because of declination, a near-universal tendency for pitch to gradually lower over the course of an utterance. Declination is indeed attested in Drubea and Numèè, where an utterance containing only registerless syllables tends to 'follow a straight-falling melodic line' (Rivierre 1973: 131).

As shown by Rivierre (1973: 125), the contrast between utterance-initial downstepped and registerless syllables is maintained in the presence of a following downstepped syllable: in this context, an initial registerless syllable undergoes pre-downstep raising and is realised higher than the baseline, while an initial downstepped syllable does not, and is realised at the baseline. This is illustrated by the contrast between utterance-initial registerless /goo/ in (32) and downstepped /[↓]goo/ in (33).

- (32) /goo ⁺mie/
 [⁺goo ⁺mie]
 [goo⁵ mie³]
 plant.sp wet
 ‘wet *Hibbertia pancheri*’ (Drubea; Rivierre 1973: 125)
- (33) /⁺goo ⁺mie/
 [(⁺)goo ⁺mie]
 [goo⁴ mie³]
 tree wet
 ‘wet tree’ (Drubea; Rivierre 1973: 125)

This partly explains why the default ceiling of the utterance-initial register is not at the highest point of the speaker's range (i.e., level 5): it must remain possible to raise the default register when pre-downstep raising affects the utterance-initial syllable.

3.6. Disyllables

Disyllabic stems can be classified into three register classes (Rivierre 1973: 124, 126–127; Shintani & Païta 1990b: 17–18).²⁴ These three classes are illustrated in (34) with a minimal triple.

- (34) a. Type 1: registerless / $\sigma \sigma$ / for example, /ku_re/ ‘forest’
 b. Type 2: downstepped initial syllable / $^{\downarrow}\sigma \sigma$ / for example, /[↓]ku_re/ ‘end’
 c. Type 3: downstepped second syllable / $\sigma^{\downarrow}\sigma$ / for example, /ku_r[↓]e/ ‘crayfish’

Registerless disyllables (type 1) behave exactly like a succession of two registerless monosyllables: they are realised at the same pitch as the preceding syllable, as with /veto/ ‘put’ in (35) and Figure 15; they may be affected by pre-downstep raising, as with /tea/ ‘to go up’ and /mwεɛ/ ‘again’ in (36), or by either of the two utterance-final phenomena described in §§3.3 and 3.4: utterance-final raising in Drubea, for example, /kapwa/ ‘corrugated iron’ in (37) and Figure 16, and utterance-final downstepping in Numèè, for example, /mwogo/ ‘alive’ in (38), where downstepping of the final syllable causes pre-downstep raising of the initial.

²⁴A fourth class, CV⁺V.CV, is also attested in recent loanwords, as discussed in §3.7.

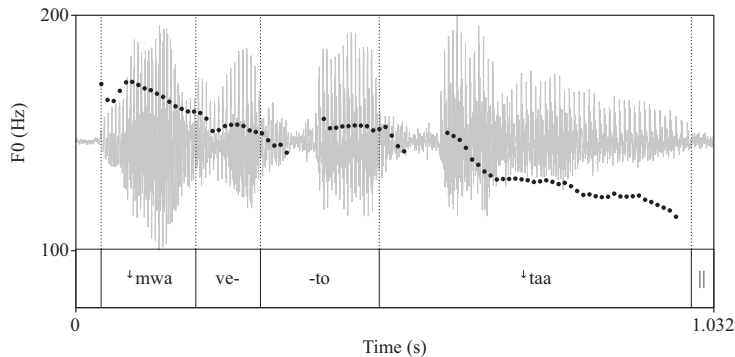


Figure 15. /...⁺mwa veto ⁺taa.../ ‘[I] put a [blanket on the table]’ (Drubea; Shintani 2019: 97, L12).

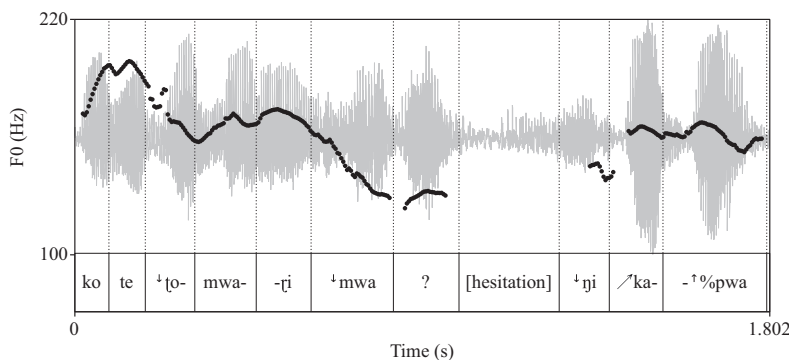


Figure 16. /ko te ⁺to-mwa^{ti} ⁺mwa... ⁺ni kapwa + ⁺%/ ‘I covered the house with corrugated iron’ (Drubea; Shintani 2019: 147, L19).

- (35) /... ⁺mwa veto ⁺taa.../
[... ⁺mwa veto ⁺taa...]
PFV put one
‘[I] put a [blanket on the table].’ (Drubea; Shintani 2019: 97, L12)
- (36) /⁺ni ⁺mwa tea mwe^{te} ⁺mee/
[⁺ni ⁺mwa te⁷a ⁷mwe^{te} ⁺mee]
[ni⁵ mwa⁴ te⁴a^{4.5} mwe⁵te⁵ mee⁴]
3SG.SBJ PFV go.up again come
‘He comes back up.’ (Drubea; Rivierre 1973: 134)
- (37) /ko te ⁺to-mwa^{ti} ⁺mwa... ⁺ni kapwa + ⁺%/
[ko te ⁺to-⁺mwa^{ti} ⁺mwa... ⁺ni ⁷ka⁺%pwa]
1SG.SBJ DESCR stay-close house with corrugated.iron
‘I covered the house with corrugated iron’ (Drubea; Shintani 2019: 147, L19)
- (38) /⁺cñbu ⁺mwã ⁺ku mwo^{to} + ⁺%/
[(⁺)cñ⁺bu ⁺mwã ⁺ku ⁺mwo⁺%to]
[cñ⁴bu⁵ mwã⁴ ku³ mwo⁴to³]
rat PFV flee alive
‘The rat escaped safe and sound.’ (Numèè; Rivierre 1973: 146)

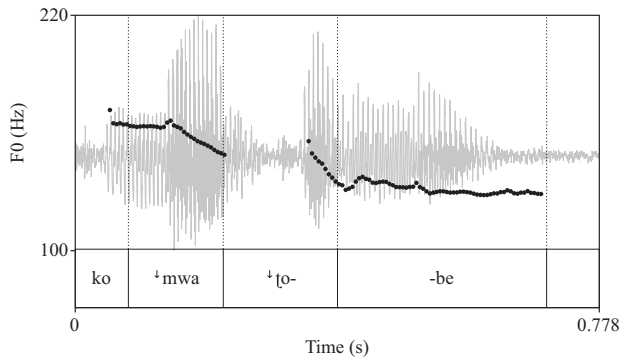


Figure 17. /ko ʔmwa ʔtobe/ ‘I woke up’ (Drubea; Shintani 2019: 193, L26).

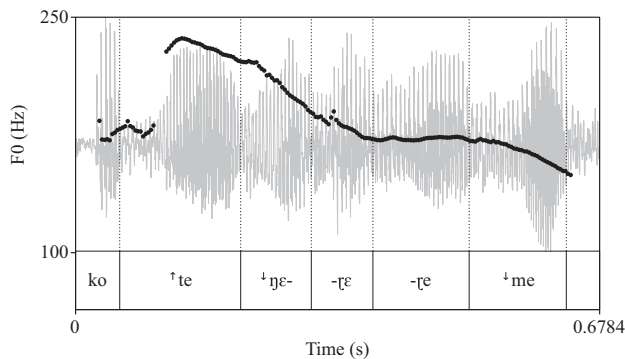


Figure 18. /ko te ʔɲɛɛ-ɾɛ ʔme.../ ‘I think that...’ (Drubea; Shintani 2019: 214, L29).

The initial syllable of a type 2 disyllable behaves like a downstepped monosyllable: it is realised lower than the preceding syllable, irrespective of whether that syllable is downstepped or registerless. In example (39), the initial downstepped syllable of the verb /ʔtobe/ ‘wake up’ is realised lower than the immediately preceding downstepped syllable /ʔmwa/, as seen in Figure 17. As with downstepped monosyllables, the first syllable of type 2 disyllables frequently causes raising of the preceding registerless syllable(s), as illustrated in example (40), where the registerless descriptive marker /te/ is clearly raised before the verb /ʔɲɛɛ-ɾɛ/ ‘think’, as shown in Figure 18 (see also /ʔbeɾu/ ‘swim’ in (13b) and /ʔkeɾɛ/ ‘eat’ in (14)).

- (39) /ko ʔmwa ʔtobe/
 [ko ʔmwa ʔtobe]
 ISG.SBJ PFV wake.up
 ‘I woke up.’ (Drubea; Shintani 2019: 193, L26)

- (40) /ko te ʔɲɛɛ-ɾɛ ʔme.../
 [ko ʔte ʔɲɛɛ-ɾɛ ʔme...]
 ISG.SBJ DESCR think-ACT that
 ‘I think that...’ (Drubea; Shintani 2019: 214, L29)

The second syllable of type 2 disyllables behaves like a registerless syllable. It is realised by default at the same pitch as the initial syllable, as in (40) and Figure 18. It is also subject to pre-downstep raising, as well as utterance-final raising in Drubea – although this is rarely the case for light syllables,

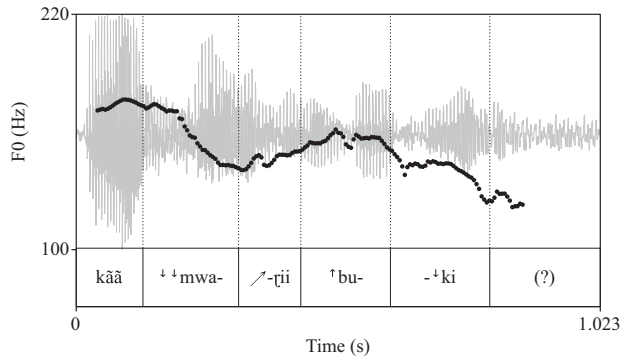


Figure 19. /kã̃*ã* mwaɽii bu*ki/ ‘[I] planted flowers’ (Drubea; Shintani 2019: 147, L19).

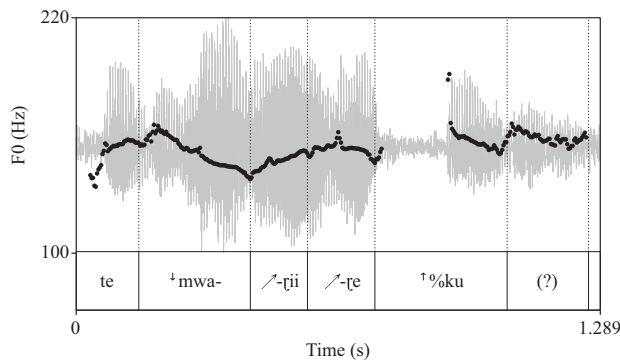


Figure 20. /te* mwaɽii-ɽe ku +*%/ ‘[I] plant yams’ (Drubea; Shintani 2019: 234, L32).

and is seen mostly with long second syllables.²⁵ This is illustrated with the verb /^{*}mwaɽii/ ‘plant’ in Drubea, affected by pre-downstep raising in (41) and Figure 19 and by utterance-final raising in (42) and Figure 20.²⁶

- (41) /kã̃*ã* mwaɽii bu*ki/
 [kã̃*ã* mwa/ɽii* bu*ki]
 PROS plant flower
 ‘[I] planted flowers.’ (Drubea; Shintani 2019: 147, L19)

- (42) /te* mwaɽii-ɽe ku +*%/
 [te* mwa/ɽii-/ɽe* %ku]
 DESCR plant-ACT yam
 ‘[I] plant yams.’ (Drubea; Shintani 2019: 234, L32)

Finally, in type 3 disyllables, the second syllable is always realised at a lower pitch than the first one, just as a downstepped monosyllable would be. This is illustrated with the disyllable /ve^{*}yuu/ ‘to be sick, to die’ in (43) and (44). The pitch drop affecting the second syllable is clearly visible in Figures 21 and 22. As for the initial syllable, it behaves exactly like a registerless monosyllable: it either is realised

²⁵Both Rivierre (1973: 124, 127) and Shintani & Païta (1990b: 18) note a strong tendency to realise the long vowel in the second syllable of a type-2 disyllable (^{*}CV.CV:) with raised pitch when pronounced in isolation.

²⁶The syllable-internal downstep in /kã̃*ã/ and its realisation as a double downstep on the following syllable is discussed in §3.7. In both examples, the final syllable in Shintani’s transcription seems to be followed by a faint, nasal-sounding element. It is not clear whether this is an actual morpheme not reported in the transcription, or meaningless nasalised lengthening of the final vowel typical of utterance-final laxing, or something else. This is represented as ‘(?)’ in Figures 19 and 20.

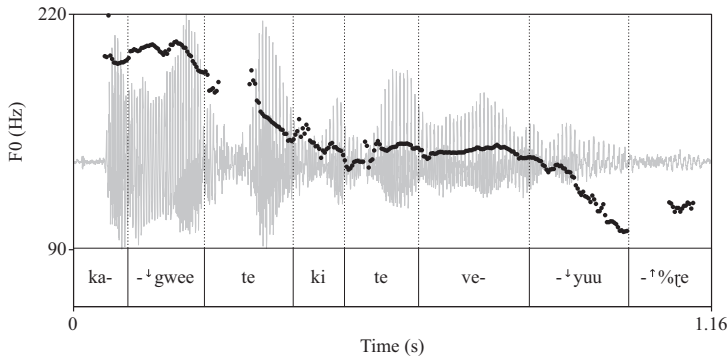


Figure 21. /... ka⁺gwee te ki te ve⁺yuu-ɽe/ ‘You look sick’ (lit. ‘it is like you are sick’; Drubea; Shintani 2019: 130, L17).

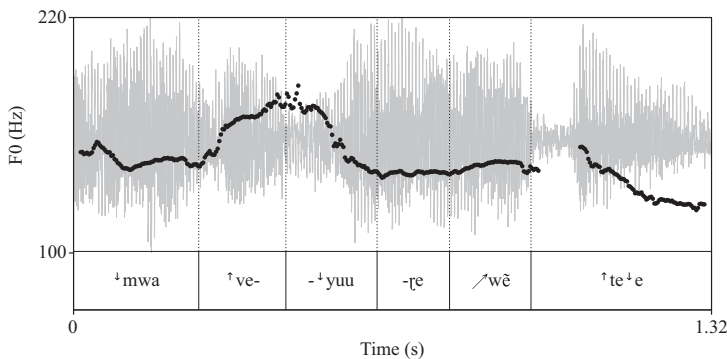


Figure 22. /... ⁱmwa ve⁺yuu wẽ te⁺e.../ ‘... [they] died because...’ (Drubea; Shintani 2019: 235, L32).

at the same pitch as the preceding syllable, as in (43) and Figure 21, or undergoes pre-downstep raising, as in (44) and Figure 22.

- (43) /ka⁺gwee te ki te ve⁺yuu-ɽe +⁺%/
 [ka⁺gwee te ki te ve⁺yuu-⁺%ɽe]
 like DET 2SG.SBJ DESCR be.sick-ACT
 ‘You look sick.’ (lit. ‘it is like you are sick’; Drubea; Shintani 2019: 130, L17)

- (44) /... ⁱɲi ⁱmwa ve⁺yuu-ɽe wẽ te⁺e ⁱɲi ⁱmwa ɲimie-⁺kāã/
 [... ⁱɲi ⁱmwa ⁱve⁺yuu-ɽe /wẽ ⁱte⁺e ⁱɲi ⁱmwa /ɲimi⁺e-⁺kāã]
 3PL.SBJ PFV die-ACT because 3PL.SBJ PFV get.old
 ‘[Some others,] they died of old age.’ (lit. ‘they died because they grew old’; Drubea; Shintani 2019: 235, L32)

3.7. CV⁺V syllables

In addition to registerless CV(V) and downstepped ⁺CV(V) syllables, there is a third prosodic type which deserves attention: syllables with a long vocalic nucleus whose second mora is downstepped, that is, (C)V⁺V. The three-way contrast is firmly established in both languages, where minimal triplets are frequent. A few examples from Drubea are listed in (45).

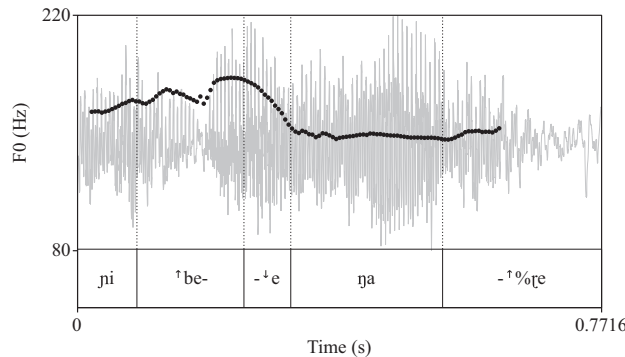


Figure 23. /*ni be⁺e ŋa-re*/ ‘[He said that] he doesn’t work’ (Drubea; Shintani 2019: 67, L8).

(45) *Drubea* (Shintani & Païta 1990b: 19)

- | | | |
|----|----------------------|--|
| a. | /be ⁺ e/ | NEGATION |
| | /bee/ | ‘fish’ |
| | / ⁺ bee | ‘descendance’ |
| b. | /pwa ⁺ a/ | ‘group of men’ |
| | /pwaa/ | ‘white’ |
| | / ⁺ pwaa/ | ‘packet’ |
| c. | /ko ⁺ o/ | ‘place, field’ |
| | /koo/ | ‘fish sp.’ |
| | / ⁺ koo/ | ‘egg’ |
| d. | /ke ⁺ e/ | IPL.SBJ |
| | /kee/ | ‘husband’ |
| | / ⁺ kee/ | ‘ <i>Broussonetia papyrifera</i> , paper mulberry’ |
| e. | /kã ⁺ ã/ | PROSECTIVE |
| | /kãã/ | ‘parent, friend’ |
| | / ⁺ kãã/ | ‘big’ |

The prosodic nature and behaviour of CV⁺V syllables is illustrated in (46) with the negative marker /be⁺e/. As can be seen in Figure 23, the second mora is downstepped; that is, it is realised within a lower register than the preceding mora and imposes this lower register on the following registerless syllable. In most cases, the initial mora of CV⁺V syllables is realised with pre-downstep raising, which is seen in the realisation [⁺be⁺e] of the negative marker /be⁺e/ in (46) and Figure 23, and more clearly in (47) and Figure 24.²⁷ However, this is not always the case, as shown in (48) and Figure 25, where the initial mora of [be⁺e] is clearly realised at the same pitch as the preceding subject pronoun [(⁺)ŋi].²⁸

- (46) /*ni be⁺e ŋa-re + ⁺%*/
- [*ni ⁺be⁺e ŋa-⁺%re]*
- 3SG.SBJ NEG work-ACT
- ‘[He said that] he doesn’t work.’ (Drubea; Shintani 2019: 67, L8)

²⁷In Figure 24, the pitch starts rather low and gradually reaches the expected utterance-initial baseline late, at the very end of the first syllable, possibly because of the initial nasal in [(⁺)ŋi] (nasal stops have a tendency to lower the pitch in Drubea). The downstep of the second syllable /⁺mwa/, is, in turn, delayed to the end of the syllable.

²⁸The final syllable [re] in Figure 25 is almost whispered, and its pitch is not detected by Praat, but it sounds the same as that of the preceding syllable [kwe].

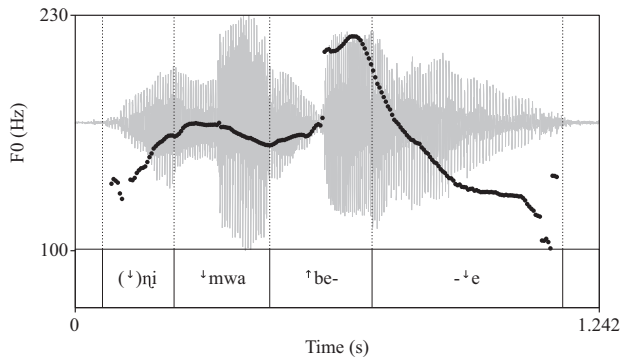


Figure 24. /ʔni ˥mwa be˥e || .../ ‘They don’t [think about working]’ (Drubea; Shintani 2019: 306, L43).

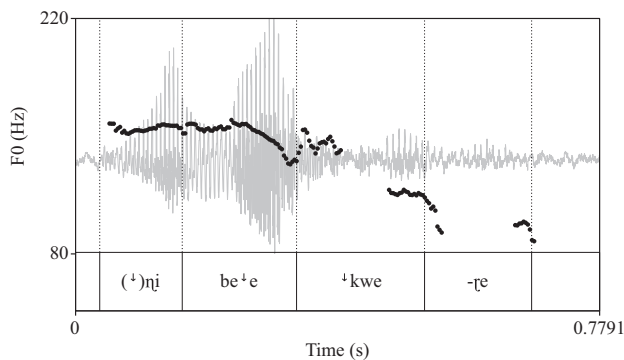


Figure 25. /ʔni be˥e kwe-ɽe/ ‘They don’t eat [it]’ (Drubea; Shintani 2019: 187, L25).

- (47) /ʔni ˥mwa be˥e || .../
 [(ʔni ˥mwa ˥be˥e || ...]
 3PL.SBJ PFV NEG
 ‘They don’t [think about working].’ (Drubea; Shintani 2019: 306, L43)

- (48) /ʔni be˥e ˥kwe-ɽe/
 [(ʔni be˥e ˥kwe-ɽe]
 3PL.SBJ NEG eat-ACT
 ‘They don’t eat [it].’ (Drubea; Shintani 2019: 187, L25)

The downstep affecting the second mora is frequently displaced and realised on the following syllable. If that syllable is underlyingly registerless, it is then realised with a downstep; that is, /CV˥V CV.../ → [CVV ˥CV...]. This is shown in (49) and Figure 26, and in (50), where the downsteps in the negative marker /be˥e/ and the quantifier /pwa˥a/ are displaced and realised on the initial syllable of the following word. Note that pre-downstep raising also applies in /be˥e/ = [˥bee ˥...] and /pwa˥a/ = [˥pwa˥a ˥...].

- (49) /ko be˥e jaanɽi-ɽe ˥me tu˥mwa auɽe ˥ne ˥mee .../
 [ko ˥bee ˥jaa/ɽi-˥ɽe ˥me ˥tu˥mwa auɽe ˥ne ˥mee ...]
 1SG.SBJ NEG want-ACT that other people FUT come
 ‘I do not want other people to come [help me plant yams].’ (Drubea; Shintani 2019: 48, L05)

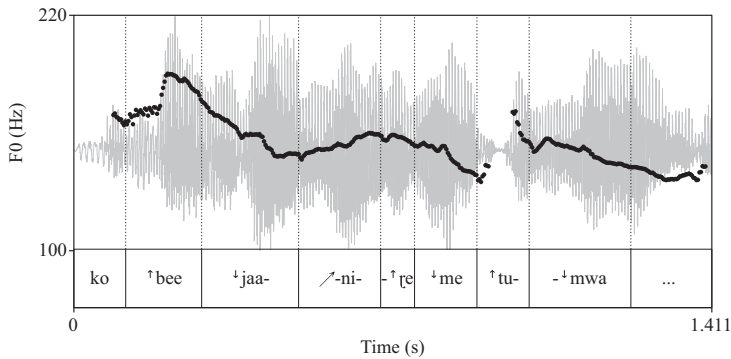


Figure 26. /ko be⁺e jaani-ɽe ⁺me tu⁺mwa .../ ‘I do not want other [people to come help me plant yams]’ (Drubea; Shintani 2019: 48, L05).

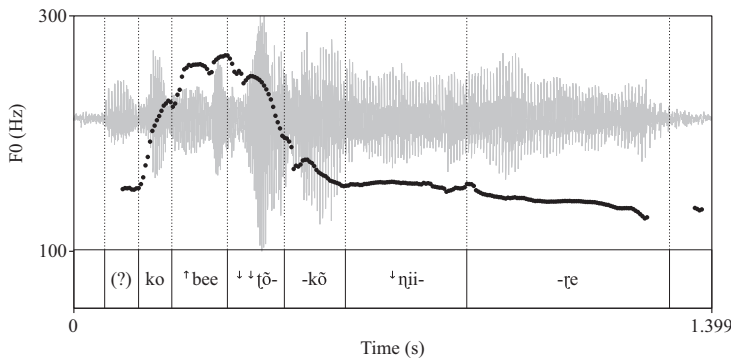


Figure 27. /ko be⁺e ⁺tōkō ⁺nii-ɽe .../ ‘I cannot say...’ (Drubea; Shintani 2019: 67, L08).

- (50) /kḁḁ pwa⁺a ɽe a ɽe⁺ɛ/
 [kḁḁ ⁺pwa⁺a ⁺ɽe a ⁺ɽe⁺ɛ]
 [kḁḁ⁴ pwa⁵ ɽe⁴ a⁴ ɽe⁵ɛ³]
 one.of all ART REL Treē
 ‘one man from the Treē clan’ (Drubea; Rivierre 1973: 143)

When the following syllable is underlyingly downstepped, downstep displacement creates a double downstep; that is, /CV⁺V ⁺CV.../ → [CVV ⁺⁺CV...], as illustrated by /be⁺e ⁺tōkō/ [bee ⁺⁺tōkō] in (51) and Figure 27, and by /ya⁺a ⁺me/ [yaa ⁺⁺me] and /ge⁺e ⁺me/ [gee ⁺⁺me] in (52).

- (51) /ko be⁺e ⁺tōkō ⁺nii-ɽe .../
 [ko ⁺bee ⁺⁺tōkō ⁺nii-ɽe ...]
 1SG.SBJ NEG be.able say-ACT
 ‘I cannot say...’ (Drubea; Shintani 2019: 67, L08)
- (52) /ya⁺a ⁺me ge⁺e ⁺me na⁺i/
 [⁺yaa ⁺⁺me ⁺gee ⁺⁺me ⁺na⁺i]
 [yaa⁵ me³ gee⁵ me³ na⁴i³]
 NEG that IPL.EXCL.SBJ FUT arrive
 ‘We will not arrive.’ (Numèè; Rivierre 1973: 143)

That there is indeed a double downstep in such cases is confirmed by the fact that the contrast between registerless and downstepped syllables is maintained under downstep displacement, as illustrated by the

near-minimal pair in (53): underlyingly downstepped /⁺kwe-ɾe/ ‘eat’ in (53b) is realised with a starker pitch drop under downstep displacement than registerless /kwe-ɾe/ ‘dance’ in (53a), which corresponds to the cumulative effect of its underlying downstep and the displaced downstep from preceding /ko⁺o/.

(53) *Drubea* (Rivierre 1973: 144)

- | | |
|--|---|
| <p>a. /ko⁺o kwe-ɾe/
 [↑koo⁺ kwe-ɾe]
 [koo⁵ kwe⁴-ɾe]
 place dance-ACT
 ‘place of dancing’</p> | <p>b. /ko⁺o⁺ ⁺kwe-ɾe/
 [↑koo⁺ ⁺kwe-ɾe]
 [koo⁵ kwe³-ɾe]
 place eat-ACT
 ‘place of eating’</p> |
|--|---|

CV⁺V syllables are attested in both monosyllables and disyllables. The latter are exclusively recent loanwords from French and English, in which the downstep is always in the initial syllable; that is, they all have the shape CV⁺V.CV(V). A few *Drubea* examples are given in (54).

- (54)
- | | | |
|--------------------------|-----------------|-----------------------------|
| a. /o ⁺ oci/ | ‘horse’ | < English |
| b. /pi ⁺ iki/ | ‘pig’ | < English |
| c. /po ⁺ oci/ | ‘pouch, pocket’ | < French <i>poche</i> |
| d. /ta ⁺ aci/ | ‘bowl’ | < French <i>tasse</i> ‘cup’ |
| e. /co ⁺ oɾo/ | ‘salt’ | < English |
| f. /ka ⁺ aɾe/ | ‘car’ | < English <i>cart</i> |

As seen above, CV⁺V sequences behave differently from CV.⁺CV disyllables, with which there is no downstep displacement. It is not entirely clear whether CV.⁺V disyllabic words (with two different vowels) behave similarly to or differently from CV⁺V syllables. These are analysed as involving two syllables by Rivierre, an analysis I do not question in this article. There are only nine such words in Shintani & Païta’s (1990a) *Drubea* dictionary, many of which are low-frequency words (e.g., specific worm or shellfish species). The only two that are found with mid-to-high frequency in Shintani’s recordings (Shintani 2019; Académie des Langues Kanak 2023) are the verbs /u⁺i/ ‘to do, to make’ and /ya⁺i/ ‘to arrive’. I have not found one clear instance of downstep displacement involving these two words in the corpus. The speech rate of the speakers is too high in most cases to make it possible to clearly determine the exact realisation, and in those rare cases of relatively careful speech, the downstep is not displaced, but clearly realised on the second vowel: [u⁺i], [ya⁺i]. The Numèè cognate of the verb ‘to arrive’ is /pa⁺i/. This verb is used several times in every one of the eight Numèè texts collected by Haudricourt & Rivierre (n.d.) and archived in the Pangloss collection. In all instances, the downstep is clearly heard between the two vowels, with no displacement. This is not a full and detailed investigation of this question, for which more carefully articulated data are necessary, but it seems to at least suggest that Rivierre was likely correct in identifying CV⁺V syllables as having special status, different from CV.⁺V sequences (and more generally that his analysis of sequences of like vowels as long vowels and sequences of unlike vowels as heterosyllabic is likely correct).

Note that the existence of CV⁺V syllables shows that the element carrying the indication of register is not the syllable, but the mora. I will come back to this in detail in §4.2.

3.8. Stems of more than two syllables

While most stems in *Drubea* and Numèè are either mono- or disyllabic, stems of more than two syllables (mostly three or four) are also attested. However, neither Rivierre (1973) nor Shintani & Païta (1990b) give a detailed description of their tonal behaviour. Rivierre acknowledges the difficulty of establishing the morphological structure of words of more than two syllables. Compounding is indeed frequent, and many of the long words that he was not able to identify as compounds may turn out to be complex,

something he was not able to systematically verify in the field for lack of time (Rivierre 1973: 129–130).

On the basis of the limited data he collected, Rivierre (1973: 130) hypothesizes that tri- and tetrasyllables, like mono- and disyllables (cf. §3.10 below), are limited to containing at most one downstep ('L tone' in his analysis).²⁹ Rivierre further surmises that those with more than one downstep are actually compounds. This implies that each member of a compound constitutes an independent prosodic domain (which I call 'stem' here). The culminativity constraint of downstep is enforced within this domain, but not at the level of the entire morphological word. Indeed, it is quite common for compounds to include more than one downstep when they combine several stems containing a downstepped element, as in the examples in (55) (compound elements are separated by a middle dot).

- (55) a. /ko⁺o⁺·mwa/
 field·house
 'mound (on which houses are built)'
 b. /⁺ʊʊ·ũ·poo⁺ka/
 mat·hair·animal
 'blanket'
 c. /⁺kɔ⁺·vuu-ɾe/
 manner·speak-ACT
 'voice, language'

Rivierre's hypothesis is supported by the fact that, cross-linguistically, compounds have a tendency not to constitute a single tonal or prosodic domain. Examples include such different languages as the isolate Laal of Chad (Lionnet 2022a) and the Jul'hoan language of Namibia (Kx'a, formerly Northern Khoisan; Miller 2010). Closer to Drubea and Numèè, this is also the case in Paicî, another tonal language of New Caledonia, in which each member of a compound constitutes an independent tonal domain, notably for the application of a similar culminativity constraint limiting the number of downsteps per tonal domain to exactly one (Lionnet 2022b).

I set tri- and tetrasyllabic words aside in this article. A full investigation of their prosodic behaviour requires additional data collection from native speakers, which I have not been able to conduct yet. I can only say that in transcribing a selection of Shintani's (2019) recorded texts, none of the (seemingly) monomorphemic tri- or tetrasyllabic words I have encountered contained more than one downstepped syllable, or jeopardised in any way the register analysis proposed in this article.

3.9. *Verbal classifier prefixes*

Morpheme concatenation in Drubea and Numèè, either by compounding or by affixation, does not seem to interact in any significant way with the prosodic behaviour of the morphemes involved. Affixes are all registerless, except verbal classifier prefixes.³⁰ These modify the meanings of verb bases in systematic ways, such as /xa-/ 'VERB with one's teeth', /gi-/ 'VERB with a knife'. These prefixes are always monosyllabic, and take on the prosodic specification of the initial syllable of the verb they are prefixed to (Rivierre 1973: 140–141; see also Shintani & Païta 1990b: 46). This is illustrated with the Drubea prefix /ʈa-/ 'VERB with one's hand' in (56).

²⁹Shintani & Païta (1990b: 21–22) come to the same conclusion, although they note that tri- and tetrasyllables fall into several classes depending on what syllable carries the downstep.

³⁰Both Rivierre (1973: 138–140) and Shintani & Païta (1990b) analyse all affixes except the verbal classifier prefixes as toneless. Rivierre's description of the contextual realisations of the possessive suffix /-ɾe/ ~ /-V/ and the active voice suffix /-ɾe/ seems to indicate a difference in behaviour between these suffixes and what he analyses as H-toned syllables (= registerless in my analysis). It is not clear to me that this distinction is warranted, since most of the contextual realisations described by Rivierre are fully compatible with a registerless analysis of these morphemes under the analysis proposed here, and the few remaining doubts about this compatibility cannot be cleared with the data presented by Rivierre. I will consider that suffixes are registerless, simply noting that more work is necessary to confirm this.

- (56) a. /ʈa-uʀu/ ʈa-uʀu ‘cut by hand’
b. /ʈa-⁺tie/ ⁺ʈa-⁺tie ‘tear by hand’

The example in (57) shows that the prefix in /ʈa-⁺tie/ [⁺ʈa-⁺tie] is indeed downstepped, as can be seen from the fact that it triggers pre-downstep raising of the preceding registerless syllable [⁺te]. The fact that there is still a pitch drop between the prefix and the verb also shows that the downstep on the prefix is indeed a copy of the downstep of the initial syllable of the verb /⁺tie/ rather than a realignment of this downstep to the left edge of the morphological word.

- (57) /ko te ʈa- ⁺tie -ʀe/
ko te ⁺ʈa- ⁺tie -ʀe (downstep copy)
[ko ⁺te ⁺ʈa- ⁺tie -ʀe]
[ko⁴ te^{4.5} ʈa⁴ ti³e³ -ʀe³]
ISG.SBJ DESCR with.hand- tear -ACT
‘I tear by hand.’ (Drubea; Rivierre 1973: 141)

3.10. Summary

The word-prosodic system of Drubea and Numèè is entirely built on one underlying binary contrast: registerless (i.e., prosodically unspecified) vs. downstepped elements. These elements are either entire syllables as in CV(V) and ⁺CV(V), or the second mora of a syllable, as in the case of CV⁺V syllables, which suggests that both the syllable and the mora have a role to play, a point I address in more detail in §4.2. Downstep is realised as a downward register contrast with what precedes. It is not realised utterance-initially, where the conditions for this contrast are not met. In that case, the downstep is simply left unrealised, and downstepped elements are pronounced just like registerless ones at baseline pitch, a high-mid default pitch schematically represented as level 4 on the typical 1-to-5 scale. Registerless elements are prosodically inert; that is, their realisation is entirely context-dependent. When utterance-initial, they are realised at baseline pitch. In non-initial position, they are realised by default at the same pitch as the preceding (downstepped or registerless) element. When followed by a downstepped element (including when in utterance-initial position), they tend to undergo pitch raising, to emphasize the downward contrast that follows. Finally, registerless elements are sensitive to utterance-final prosodic phenomena: final raising in Drubea, final lowering in Numèè.

The distribution of downstep within mono- and disyllabic stems is summarised in Table 2. One interesting property that emerges is CULMINATIVITY: a stem may include at most one downstep, as was already noted by Rivierre (1973: 128).³¹ This downstep may affect the first or second syllable of disyllables. It may also affect the second mora of a bimoraic CVV syllable. In the native vocabulary, this is found only with monosyllabic CV⁺V stems. CV⁺V.CV(V) stems are also attested, but all are

Table 2. Downstep distribution in mono- and disyllabic stems.

Registerless	σ1 downstepped	σ2 downstepped	Intra-syllabic downstep
CV	⁺ CV	—	—
CVV	⁺ CVV	—	CV ⁺ V
CV.CV	⁺ CV.CV	CV. ⁺ CV	—
CV.CVV	⁺ CV.CVV	CV. ⁺ CVV	—
CVV.CV	⁺ CVV.CV	CVV. ⁺ CV	CV ⁺ V.CV (loans)
CVV.CVV	⁺ CVV.CVV	CVV. ⁺ CVV	CV ⁺ V.CVV (loans)

³¹Shintani & Païta (1990b) do not explicitly make this claim, but their sketch of the prosodic system is compatible with it.

recent loanwords, mostly from English and French. CV(V).CV⁺V stems are strictly unattested. (The optionality of the onset consonant is not indicated in the schematic syllable structures given in Table 2; CV, CVV and CV(V) stand for (C)V, (C)VV and (C)V(V), respectively.)

4. Register analysis

In the preceding section, I described the word-prosodic system of Drubea and Numèè as consisting of a contrast between downstepped units and registerless units. In this section, I propose an analysis of this contrast and of the prosodic behaviour of these units in terms of register features, demonstrating that no tone or tonal feature is necessary to account for the word-prosodic system of Numèè and Drubea, which are thus not tonal languages *stricto sensu*, but languages that use register for lexical contrast.

4.1. A brief history of the representation of tone and register

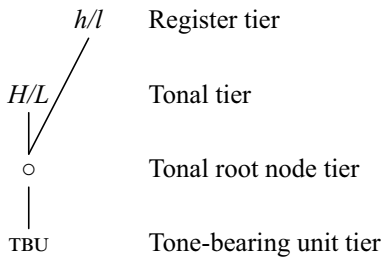
The representation of register phenomena like downstep and upstep results from a long history of attempting to extend featural representations to tone – that is, to represent tone as a bundle of features, and explain tonal behaviour with the properties of these features (e.g., Wang 1967; Yip 1980, 1989; Clements 1981; Hyman 1993; Pulleyblank 1986; Snider 1999, 2020). The most widely used feature system is that proposed by Yip (1980) and refined by Pulleyblank (1986), which makes use of two features: one ‘register’ feature [±upper], dividing the tone range into an upper and a lower register, and a secondary feature [±high] (Yip) or [±raised] (Pulleyblank) further subdividing each register into two discrete tonal categories. In this system, a four-height tonal contrast would be analysed as in (58) below.³²

(58)	Tone height	Possible transcription	[upper]	[high]/[raised]
	4	ǎ	+	+
	3	á	+	–
	2	ā	–	+
	1	à	–	–

Snider’s (1999; 2020) Register Tier Theory is superficially very similar to the above approaches in proposing a four-way contrast using two types of features: two REGISTER FEATURES *h* (high) and *l* (low), and two TONE FEATURES *H* (High) and *L* (Low). Snider’s system is, however, different from all others in one crucial point: the definition of the register features. Indeed, while his two tone features *H* and *L* can be considered the exact unary equivalents of Yip’s and Pulleyblank’s binary [+raised]/[+high] and [–raised]/[–high], respectively, his two register features *h* and *l* are not equivalent to Yip’s and Pulleyblank’s register feature [±upper]. The latter is defined in purely paradigmatic terms: [+upper] tones are realised within the upper half of the register, while [–upper] tones in the same environment would be realised in the lower half of the *same register*. In contrast, Snider’s *h* and *l* register features are defined in syntagmatic terms: they ‘effect a register shift *h* = higher and *l* = lower *relative to the preceding register setting*’ (Snider 2020: 25; emphasis mine; see also pp. 151–153). In other words, *h* and *l* are defined as overt representations of upstep (*h* = ↑) and downstep (*l* = ↓), which I show in this section are very well suited to account for the Drubea and Numèè facts described above.³³

Snider (1999, 2020) further refines the featural representation of tone by proposing a geometry in which register and tone features are linked to a tonal root node (TRN), which is itself associated with a tone-bearing unit (TBU), as shown in (59).

³²Other authors have proposed very similar systems with unary rather than binary features (e.g., Clements 1981; Hyman 1993).
³³For earlier proposals on the representation of downstep and upstep, see Clements (1979, 1981); Hyman (1985b, 1993); Inkelas *et al.* (1987); Inkelas (1989).

(59) *Geometry of tone* (Snider 2020: 23)

The TRN unites register and tone features into a single bundle defining individual tones (e.g., low = *Ll*, high = *Hh*, mid = *Lh* or *Hl*), exactly as the C and V root nodes in feature geometry unite segmental features into bundles defining individual segments (Sagey 1986; Clements & Hume 1995). This representation makes it possible to account for the fact that low tones have a very strong cross-linguistic tendency to downstep a following H tone. Indeed, a low tone is always specified as *l* on the register tier (and *L* on the tone tier), that is, as a downstep-inducing element.³⁴

In the rest of this section, I show that register features, in Snider's syntagmatic definition, are sufficient to account for the word-prosodic system of Drubea and Numèè. In other words, register features need not be paired with tone features and may be active in a phonological system even in the absence of tone features.

4.2. Stem-level patterns and the register-bearing unit

I propose to analyse downstepped RBUs in Drubea and Numèè as being underlyingly associated with a *l* register feature. Note that in the absence of tone features, the TRN is unnecessary, and so I omit it from the representations. The distribution of this *l* feature in mono- and disyllabic stems, summarised in Table 2, has two characteristics: (i) there are only three stem-level register patterns: $\emptyset$, *l*, and $\emptyset l$; and (ii) the RBU is the mora, with a strong syllabic constraint: register features associate with the leftmost mora of a syllable. This accounts for all the native mono- and disyllabic stem shapes described in §3 and listed in Table 2, that is, all register-less stems (the $\emptyset$ pattern), stems with an initial downstepped syllable (the *l* pattern), and stems with a downstepped second syllable (the $\emptyset l$ pattern). It also accounts for monosyllabic CV⁺V stems, as discussed below.

This analysis predicts that syllable-internal downstep (i.e., downstep affecting the second mora of a long vowel) should only be allowed in monosyllabic bimoraic stems, as a result of their combination with the $\emptyset l$ pattern. The *l* feature in this case exceptionally associates with the second mora, for lack of a second syllable. In disyllabic stems, on the other hand there are always enough syllables for all patterns to align their *l* feature with the leftmost mora of a syllable. As can be seen in Table 3, the constraint that *l* be aligned with the left edge of the syllable means that only mappings (a) (for *l* associated with a CVV.CV(V) stem) and (c) (for $\emptyset l$ associated with a CV(V).CVV stem) are possible. As we saw in §3.10, mapping (d) is strictly unattested, and mapping (b) is found only in recent loanwords. I will ignore these loanwords in the remainder of this article. Nothing in their behaviour suggests any incompatibility with the register analysis proposed here. These could either be analysed as involving an exceptional $\emptyset l \emptyset$ pattern confined to loanwords, or be treated as compounds, that is, two separate prosodic domains: $\emptyset l$ monosyllabic CV⁺V plus register-less monosyllabic CV(V).³⁵ Note that CV⁺V syllables constitute the only evidence that the mora plays any role in register feature association. Without these stems, the RBU would be defined as the syllable. All attested native mono- and disyllabic stem shapes are thus accounted for by the analysis, as shown in Table 4.

³⁴See Snider (2020: 21–65) for a detailed description of the theory and its explanatory potential.

³⁵Similar analyses of phonotactically odd loanwords as compounds have been proposed in other languages, including Jul'hoan (Miller 2010).

Table 3. *Attested and unattested register pattern mappings on disyllables.*

	<i>l</i> pattern	$\emptyset l$ pattern
a. $\begin{array}{c} l \\ \\ \text{CVV.CV(V)} \end{array}$	✓	*
b. $\begin{array}{c} l \\ \\ \text{CVV.CV(V)} \end{array}$	*	*
c. $\begin{array}{c} l \\ \\ \text{CV(V).CVV} \end{array}$	*	✓
d. $\begin{array}{c} l \\ \\ \text{CV(V).CVV} \end{array}$	*	*

Table 4. *Mapping of register patterns onto mono- and disyllabic native stems.*

	$\emptyset$ pattern	<i>l</i> pattern	$\emptyset l$ pattern
One-syllable stems		$\begin{array}{c} l \\ \\ \text{CV(V)} \end{array}$	$\begin{array}{c} l \\ \\ \text{CVV} \end{array}$
Two-syllable stems	CV(V).CV(V)	$\begin{array}{c} l \\ \\ \text{CV(V).CV(V)} \end{array}$	$\begin{array}{c} l \\ \\ \text{CV(V).CV(V)} \end{array}$

4.3. Register analysis: the basics

An utterance containing a sequence of downstepped monosyllabic morphemes, as in (11), is represented as in (60).³⁶

(60) *Drubea* /⁺ŋi ⁺mwa ⁺ŋii ⁺me/ → [(⁺)ŋi ⁺mwa ⁺ŋii ⁺me]; cf. (11)

a. $\begin{array}{cccc} l & l & l & l \\ | & | & | & | \\ \eta i & mwa & \eta ii & me \end{array} \rightarrow$ b. $\begin{array}{cccc} (l) & l & l & l \\ | & | & | & | \\ \eta i & mwa & \eta ii & me \end{array}$

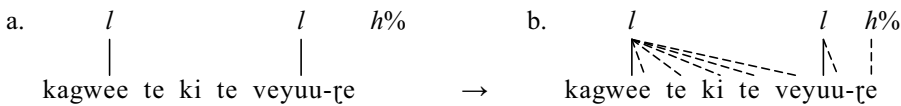
The utterance-initial downstep on /⁺ŋi/ is not realised, as seen in §3.5. We will see in §4.5 that this unrealised downstep is actually still phonologically active, and not deleted, hence its presence in parentheses in the surface transcription. Each one of the following *l* features is realised as a register drop, taking the speaker step by step all the way down to the lower end of their pitch range.

³⁶I assume that an utterance starts at a default register, which I call the ‘baseline’ (see §3.5), and which I do not transcribe. Cf. Snider (2020: 25) for a similar idea about the interpretation of utterance-initial register features.

An utterance involving only registerless morphemes remains devoid of any register indication, and is thus, by default, realised at the baseline. The same is true of utterances whose only *l*-bearing RBU is utterance-initial, as in example (31): the initial downstep is not realised, making such utterances prosodically identical to ones with no downstep at all (although see §4.5).

Registerless RBUs following a downstepped RBU are by default (and in the absence of any following downstep) realised at the same pitch as the downstepped RBU in question, as we saw in §§3.1 and 3.2. This could be interpreted as the result of spreading the *l* feature to all following registerless RBUs, as in (61), which shows the representation posited for (43); the same spreading also applies to /⁺ɲii/ in (60b). The final *h*% boundary feature will be discussed in §4.8.

(61) *Drubea* /ka⁺gwee te ki te ve⁺yuu-ɾe + ⁺%/ → [ka⁺gwee te ki te ve⁺yuu-⁺%ɾe]; cf. (43)



However, the realisation of registerless RBUs is highly variable, as we have seen. In addition to their utterance-initial default baseline realisation, they are also subject to pre-downstep raising, final raising (*Drubea*) or lowering (*Numèè*), and extrapolation. Spreading of the preceding register feature to account for same-pitch realisation is thus at best optional (if even necessary), and will henceforth not be represented. Registerless RBUs will be left unassociated, which is an apt representation of their availability to variable realisations.

4.4. Pre-downstep raising as *h*-epenthesis

Pre-downstep raising can be analysed as the optional postlexical assignment of a *h* register-raising (= upstep) feature, as illustrated in (62), where every registerless syllable followed by a downstep is affected.

(62) a. /gu⁺cape⁺panaa⁺kɔ⁺ɲu⁺wii to/
 [⁺gu⁺ca⁺pɛ⁺pa⁺ɲaa⁺kɔ⁺ɲu⁺wii to]
 [gu⁵ca⁴pɛ^{4.5}pa³ɲaa^{3.5}kɔ²ɲu^{2.5}wii^{1.5}to^{1.5}]
 2SG.SBJ raise mast on boat down there
 ‘Raise the mast on the boat!’ (*Numèè*; Rivierre 1973: 134)



This inserted *h* feature may not delete an underlying *l*. As we saw in §3.2, pre-downstep raising affects mostly registerless RBUs, and is rarely seen with downstepped RBUs. When it does affect a downstepped RBU, as in (18), the raising occurs after the pitch drop triggered by the downstep. That is, the postlexical insertion of the *h* feature creates a register contour $\widehat{lh}$ on a single mora, as illustrated in (63), which shows the representation of (18).

(63) *Numèè* /⁺ɲɛ⁺mwa⁺vɛ/ → [(⁺)ɲɛ⁺mwa⁺ $\widehat{a}$ ⁺vɛ]; cf. (18)



4.5. Utterance-initial downstep

As we saw in §3.5, the downstep is not realised utterance-initially, where there is no distinction in pitch between a downstepped RBU and a registerless one – if the following RBU is registerless. The contrast between downstepped and registerless syllable is, however, maintained whenever the utterance-initial syllable is followed by a downstepped RBU. In this case, a registerless syllable undergoes pre-downstep raising, while a downstepped one does not. This can easily be accounted for by positing that the *l* feature on the utterance-initial RBU is not deleted, but only left unrealised, for lack of a preceding RBU to contrast with. That is, the lack of phonetic distinction between registerless and downstepped syllables in this position is the result of the identical phonetic implementation of two otherwise contrastive phonological objects.

The presence of this unrealised *l* feature is revealed by the fact that it prevents pre-downstep raising, that is, *h*-epenthesis.³⁷ This can be seen by comparing (64) and (65), which represent the minimal pair in (32) and (33). In (64), registerless /goo/ ‘*Hibbertia pancheri*’ is targeted by *h*-epenthesis caused by the following downstepped adjective /⁺mie/. In (65), on the other hand, *l*-carrying /⁺goo/ ‘tree’ fails to undergo *h*-epenthesis in the same context.

(64) *Drubea* /goo ⁺mie/ → [⁺goo ⁺mie] = [goo⁵ mie³]; cf. (32)



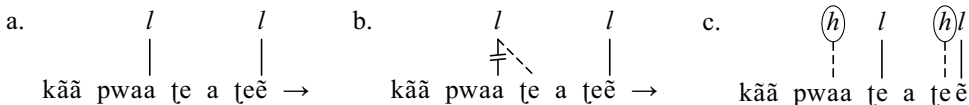
(65) *Drubea* /⁺goo ⁺mie/ → [(⁺)goo ⁺mie] = [goo⁴ mie³]; cf. (33)



4.6. CV⁺V syllables: downstep displacement and double downstep

As we saw in §3.7, the downstep in CV⁺V syllables has a tendency to be realised on the following syllable. As illustrated in (66), which shows the derivation of (50), this downstep displacement is easily represented as spreading of the *l* feature onto the next syllable followed by delinking from its original host, as with /pwa⁺a tɛ/ in (66b). This is followed by postlexical *h*-epenthesis (pre-downstep raising) in (66c).³⁸

(66) *Drubea* /kãã pwa⁺a tɛ a tɛ⁺ẽ/ → [kãã ⁺pwa⁺a ⁺tɛ a ⁺tɛ⁺ẽ]; cf. (50)

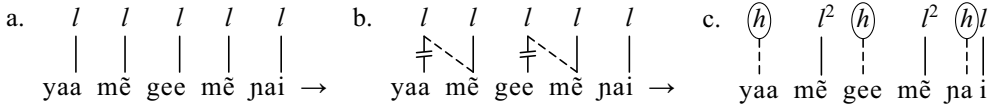


When the following syllable is already underlyingly downstepped, downstep displacement creates a double downstep, as seen in §3.7. This is illustrated in (67), which shows the derivation of (52). The *l* feature on both /ya⁺a/ and /ge⁺e/ is shifted to the next syllable in (67b), where it combines with the underlying *l* feature to produce a double downstep, represented as *l*² in (67c).

³⁷I have not seen any case of pre-downstep raising affecting an utterance-initial downstepped RBU. If this were to occur, the *lh* contour created by *h*-epenthesis, as seen in (18) and (63), would preserve the phonetic distinction between registerless and downstepped RBUs in this position.

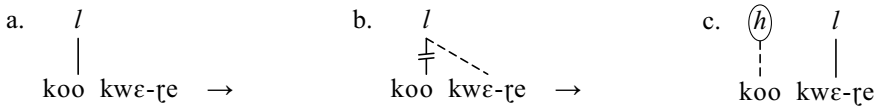
³⁸Note that /tɛ. ⁺ẽ/ is a disyllabic stem.

(67) *Numèè* /ya⁺a⁺mẽ ge⁺e⁺mẽ ja⁺i/ → [⁺yaa⁺⁺mẽ⁺gee⁺⁺mẽ⁺ja⁺i]; cf. (52)

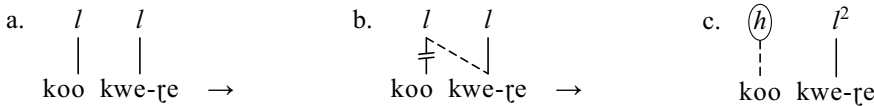


The contrast between registerless and downstepped syllables is thus maintained under downstep displacement, as seen in the near-minimal pair in (53), whose derivations are shown in (68) and (69). As can be seen, the downstep of the initial syllable /ko⁺o/ shifts to the following syllable in both cases (step (b) in both examples), and the underlying registerless-*vs.*-downstepped contrast on the following syllable (/kwɛ/ *vs.* /⁺kwe/) is maintained, only it is this time realised as a simple-*vs.*-double downstep contrast ([⁺kwɛ] *vs.* [⁺⁺kwe]; step (c)).³⁹

(68) *Drubea* /ko⁺o kwɛ-ɾɛ/ → [⁺koo⁺kwɛ-ɾɛ]; cf. (53a)



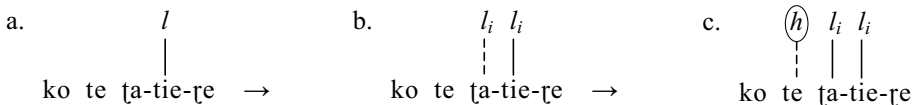
(69) *Drubea* /ko⁺o⁺kwe-ɾɛ/ → [⁺koo⁺⁺kwe-ɾɛ]; cf. (53b)



4.7. Morphological register copying

The only morphologically conditioned register operation in *Drubea* and *Numèè* is seen with verbal classifier prefixes, as discussed in §3.9. This can now be represented as copying of the *l* feature carried by the initial syllable of the verb stem to which the prefix is attached. This is illustrated in (70), which shows the derivation of (57). Register feature copying onto the prefix is shown in (70b), where the original feature and its copy are coindexed. Postlexical *h*-epenthesis then applies in (70c).

(70) *Drubea* /ko te [a-⁺tie-ɾɛ/ → [ko⁺te⁺[a-⁺tie-ɾɛ]; cf. (57)



4.8. Utterance-final prosody

I analyse utterance-final raising in *Drubea* and lowering in *Numèè*, described in §§3.3 and 3.4, as the realisation of the boundary register features *h*% and *l*%, respectively. The realisation of these boundary

³⁹Since downstep displacement appears to be by far the most frequent realisation of CV⁺V monosyllables, one may wonder whether these might be better analysed as registerless syllables followed by an underlying floating *l* feature: *includegraphicsPHO2510017 – INLINE0001 – BW.eps*. The floating downstep would be preferably realised on the following syllable, but may exceptionally be realised on its host – a case of ‘self-docking’ in Rolle’s (2018: 80–81) terms – in which case it associates only with the very last mora of the syllable. This analysis is mostly equivalent to the one I propose, with one exception: it fails to account for the absence of monomorphemic monomoraic syllables followed by a floating *l*, that is, *includegraphicsPHO2510017 – INLINE0002 – BW.eps*. This absence can easily be explained in the underlying /CV⁺V/ analysis I propose, in at least two ways: either the combination of a CV syllable with the *∅l* pattern is impossible for lack of enough morae to realise the pattern, or this combination is neutralised with CV⁺V by lengthening of the vowel to fit the pattern (/CV + *∅l* → CV⁺V). Full neutralisation is not expected in the floating *l* analysis, where one should still see a contrast between *includegraphicsPHO2510017 – INLINE0003 – BW.eps* and *includegraphicsPHO2510017 – INLINE0001 – BW.eps* when the downstep is realised on the following syllable, for example, *includegraphicsPHO2510017 – INLINE0005 – BW.eps*. Such a contrast is not attested, which militates in favour of the underlying /CV⁺V/ analysis.

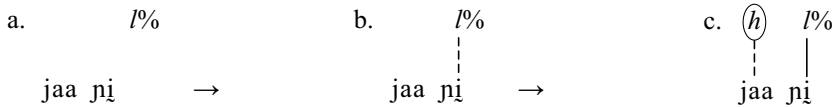
register features operates under different conditions in the two languages. In Drubea, the boundary $h\%$ (optionally) docks on any utterance-final registerless syllable, and may spread leftward onto any preceding registerless syllable. This is illustrated in (71), which shows the derivation of (20).

(71) *Drubea* / $^{\uparrow}$ taa bee pwi + $^{\uparrow}\%$ / $\rightarrow$ [$^{\uparrow}$ taa bee $^{\uparrow}\%$ pwi]; cf. (20)

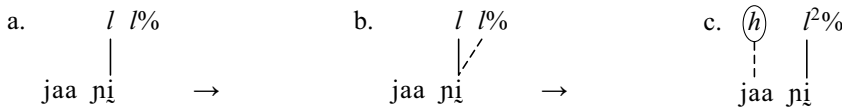


In Numèè, the boundary $l\%$ is realised only on utterance-final monomoraic syllables, both registerless and downstepped (creating a double downstep in the latter case), if they are preceded by a registerless syllable, that is, one that can undergo pre-downstep raising, as seen in §3.4. This is illustrated in (72) and (73), which show the derivations of (24) and (25).

(72) *Numèè* /jaa nĩ + $^{\uparrow}\%$ / $\rightarrow$ [$^{\uparrow}$ jaa $^{\uparrow}\%$ nĩ]; cf. (24)



(73) *Numèè* /jaa $^{\uparrow}$ nĩ + $^{\uparrow}\%$ / $\rightarrow$ [$^{\uparrow}$ jaa $^{\uparrow\uparrow}\%$ nĩ]; cf. (25)



5. Tonal alternative

In this section, I show that the register analysis proposed in §4 fares better than one in which downstep is analysed as the result of tonal interaction alone. In laying out this alternative, I ignore utterance-final prosodic effects and CV $^{\uparrow}$ V syllables, which would be straightforwardly accounted for and are not crucial for the comparison.

Rivierre's (1973) original analysis of the word-prosodic system of Drubea and Numèè is couched in purely tonal terms: the main contrast is between marked L-toned syllables (my downstepped syllables) and unmarked H-toned syllables (my registerless syllables). However, as we saw at the end of §3.1, his description of the realisation of L-toned and H-toned syllables corresponds exactly to the register analysis I propose in this article. The L tone is indeed said to be realised as a downward register contrast with the preceding syllable, that is, a downstep, while H-toned syllables are clearly described as unspecified for lexical prosody, and subject to contextual realisation (Rivierre 1973: 153). The register analysis I propose here thus builds on Rivierre's insightful description and understanding, and translates it into a more adequate framework (autosegmental phonology, downstep, register features) in order to better highlight the properties of this very unusual system and bring it to bear on contemporary phonological theory and representations.

What I would like to demonstrate here is that an analysis in purely tonal terms, that is, Rivierre's L-vs.- $\emptyset$ ~H analysis taken literally (and not in the way that Rivierre intended), would not be satisfactory.

Utterances consisting of an alternating sequence of what I analyse as registerless and downstepped syllables, as in (15) and Figure 8, are realised as a succession of peaks and valleys with a marked lowering of the pitch ceiling after every valley. This is suggestive of a much simpler analysis in terms of a succession of H and L tones, with automatic downstep of every H by the preceding L (also called downdrift; see Connell 2011): HLHLHL $\rightarrow$ HL $^{\uparrow}$ HL $^{\uparrow}$ HL = [$H^5L^3H^4L^2H^3L^1$]. That is, downstepped RBUs could alternatively be analysed as underlyingly L-toned TBUs, and registerless syllables as toneless TBUs, that is, $\emptyset$, with default H insertion on $\emptyset$ syllables to implement the underlying L-vs.- $\emptyset$ contrast.

This analysis does not, however, straightforwardly account for the behaviour of these two types of RBUs/TBUs. First, it does not naturally explain why a L-toned TBU is systematically realised lower than a preceding L-toned TBU. This behaviour is not expected of a L tone – unless one posits a postlexical Obligatory Contour Principle (OCP) constraint violated by any two successive L tones on the tonal tier at the utterance level, irrespective of morpheme and/or word boundaries, which is satisfied by inserting a downstep between every pair of adjacent L tones: LLLL → L⁺L⁺L⁺L. This is a rather unnatural and unexpected OCP constraint, which is absolutely unnecessary in the register analysis, where the downstep is not derived, but has underlying status. This OCP constraint also adds complexity to the representational apparatus, which now includes three units: underlying L, epenthetic H, and epenthetic downstep. This also complicates the analysis by recognising two types of pitch drops: the purely tonal transition from H to L (from an underlying /ØL/ sequence) and the register lowering created by downstep insertion in [L⁺L] sequences (from underlying /LL/), with no independent evidence that these are indeed two different phenomena.

The tonal analysis also fails to adequately account for the behaviour of Ø syllables. The default same-pitch realisation of Ø syllables can easily be analysed as the optional spreading of the H or L tone of the preceding syllable. However, this adds complexity to the analysis as well, since the pitch drop from the Ø TBU to the following L TBU characterising the realisation of underlying /ØL/ sequences is now analysed differently depending on the immediately preceding tone: as a [HL] sequence in /(Ø)ØL/ → [(H)HL] and as a [L⁺L] sequence in /LØL/ → [LL⁺L] (where underlining represents tone spreading).

Pre-downstep raising, in turn, would have to be analysed as involving two separate phenomena with a similar motivation and effect: raising of H immediately before L (e.g., /ØØØL/ → [HH⁺HL]), and raising of L immediately before ⁺L (e.g., /LØØL/ → [LL⁺L⁺L]), both postlexical. Raising of H immediately before L is typologically very frequent (see Hyman & Schuh 1974; Connell 2011: 12–13; and the references therein). Raising of L before ⁺L is more unexpected, but its cross-linguistic rarity is mostly due to the overall rarity of languages with downstepped L tones.⁴⁰ Interestingly, it is attested in two other tonal languages of New Caledonia, both of which have a downstepped low tone: Paicî (Lionnet 2022b: 14–15) and Cèmuhi (Rivierre 1980: 57). Note that anticipatory raising of H before downstepped ⁺L in the typical /H(L)H/ → [H⁺H] context is attested in a number of languages (see Rialland 2001 and the references therein).

To summarise, the tonal analysis suffers from both a duplication problem and a conspiracy problem. Both the downward pitch contrast characterising the so-called L-toned syllables and the optional pitch raising reinforcing this contrast are unnecessarily analysed as having two different sources, one of which – the generalised OCP-L constraint – has a strong arbitrary flavour. The register analysis, on the other hand, straightforwardly accounts for both with only one source each: underlying downstep (the *l* feature) and optional epenthetic upstep (the *h* feature), respectively. Furthermore, the tonal analysis is not representationally economic, as it cannot eschew the need to refer to downstep (and possibly upstep, depending on how raising of H before L and of L before ⁺L are analysed). The only difference with the register analysis is that downstep (and upstep) are seen as derived rather than primary/underlying. But in order to maintain that downstep (and upstep) can only be obtained through tonal interaction, the tonal analysis posits tones (underlying L and default H) for which there is actually no independent evidence. This tonal analysis – similar to Rivierre's original analysis only in that it uses the same representations, otherwise quite different, as we have seen – is thus unnecessarily convoluted, and seems to serve only one purpose: avoiding positing an underlying downstep and recognising the possibility for register features to be independent of tone and have primary/underlying status. Economy and explanatory adequacy both favour the register analysis. This analysis is all the more likely in light of the fact that underlying downsteps seem to be an areal feature of New Caledonia, where at least three other languages can arguably be analysed as having an underlying downstep: Paicî (Lionnet 2022b), Xârâcùù (Lionnet 2022b, 2025; cf. Rivierre 1978) and Cèmuhi (Rivierre 1980: 57).

⁴⁰See footnote 2 for a list of reported cases.

6. Discussion

6.1. Defining downstep

The word-prosodic system of Drubea and Numèè, with its underlying downstep and no tones, is typologically very unusual. Not only is this not a canonical case of downstep triggered by a L tone and affecting a following H tone, it is even more exceptional in not being related to tone at all. Many of the typical cross-linguistic properties of downstep are, however, related to the effect it has on (mostly H) tones. How can the contrastive pitch lowering pattern of Drubea and Numèè then be described and analysed as a form of downstep? Looking at the typical properties of downstep listed by Leben (2018: 2, building on Clements 1979, Hyman 1979 and Rialland 1997), which are quoted below in (74), (75) and (76), one can see that most are in fact found in Drubea and Numèè.

Two properties that are not found in the downstep pattern of Drubea and Numèè are the following:

(74) Leben (2018: 2):

- a. ‘Downstep preserves an affected tone’s phonological identity’; that is, ‘a downstepped High tone remains a High tone’.
- b. Downstep is phonologically distinct from a lower underlying tone, that is, a downstepped H tone is not a M tone.⁴¹

These clearly do not apply to the Drubea and Numèè prosodic system, for lack of tones. If one accepts that downstep may exist in otherwise non-tonal systems, which I hope to have demonstrated in this article, then these properties cannot be seen as crucial definitional criteria of downstep. Instead, they are the consequences of the core properties of downstep, listed in (75) below, found in phonological systems that also have tonal primitives.

The following three core properties of downstep, by contrast, are found in Drubea and Numèè, provided they are slightly rephrased:

(75) Leben (2018: 2):

- a. ‘Downstep affects not a single tone but the entire tonal sequence in its domain.’
- b. ‘Downstep’s effect is to change the register for what follows.’
- c. ‘The number of instances of downstep that can occur – in succession or in combination with other tones in an utterance – is in principle unlimited.’

Property (75a) can be said to be at work in Drubea and Numèè, if one considers that what is affected by downstep is not the ‘*tonal* sequence’, but more generally the prosodic material within its domain. In the case of Drubea and Numèè, registerless RBUs are by default realised at the same pitch height as the preceding downstepped RBU, that is, within a new, lowered register compared with the register of the RBU preceding the downstep. In any case, downstep in Drubea and Numèè does indeed ‘change the register for what follows’, as per (75b).

Property (75c), the utterance-level cumulativeness of downstep, is clearly present in Drubea and Numèè, where an utterance may be composed of a string of individual morphemes that each have one downstepped RBU. Such an utterance is realised with as many pitch drops as there are downsteps, as can be seen in (11) and in (12) and Figure 7.

The last three properties mentioned by Leben, listed in (76), are of a different nature: they are typological tendencies rather than strict definitional criteria. These should thus not be given the same importance in the definition of downstep.

⁴¹ Rephrased from ‘downstep is phonologically distinct from a Mid tone’ to accommodate languages with downstepped M and/or L.

(76) Leben (2018: 2):

- a. Utterance-initially, there is no phonetic contrast between downstep and its absence.⁴²
- b. ‘Characteristically, only High tones are contrastively downstepped, though a few cases of downstepped Low and Mid are found.’
- c. ‘Downstep functions contrastively in a variety of ways – phonological, syntactic, morphosyntactic, and lexical.’

Property (76a) is known to have a few exceptions, although it holds for a majority of languages with downstep.⁴³ We saw in §3.5 that Drubea and Numèè do conform to this typological expectation: there is indeed no phonetic difference between a downstepped and a registerless RBU in utterance-initial position (unless the following RBU is downstepped). This fact alone confirms the downstep analysis and disproves the tonal alternative entertained in the preceding section, since it is not expected for an underlying tonal contrast such as H vs. L not to be realised utterance-initially.

Property (76b) also has exceptions: 15 languages to date have been described as having a downstepped L tone, and eight languages a downstepped M tone (see footnotes 2 and 3). One of the goals of the present article is to show that, in addition to downstepped M and downstepped L, non-canonical cases of downstep include underlying downstep, that is, downstep independent of tone.

Finally, regarding the loosely defined property in (76c), one can only say that Drubea and Numèè are languages in which downstep is used exclusively for lexical contrast.

We can conclude from what precedes that the contrastive pitch drop that is the basis of the word-prosodic system of Drubea and Numèè does indeed meet what I consider to be the core definitional criteria of downstep, namely the three properties in (75), and also conforms to at least some of the cross-linguistic expectations of downstep.

6.2. Tone, register and the typology of word-prosodic systems

Given that register is not tone *stricto sensu*, and given that the register feature *l* in Drubea and Numèè is culminative (no more than one per stem), which is a property frequently associated with ‘accent’, one may hypothesize that Drubea and Numèè might be better described as accentual rather than tonal languages. I show in this section that this is not the case – although the typologically unusual word-prosodic system of Drubea and Numèè requires a refinement of the definition of what counts as a ‘tonal’ language and an enrichment of the typology of word-prosodic systems.

Hyman (2006: 225) defines two prototypical word-prosodic systems: TONE vs. STRESS ACCENT (which I simplify to ‘accent’ here). ‘Accent’ is defined as having ‘two inviolable, definitional properties: (i) obligatoriness (every word has at least one stress accent); (ii) syllable-dependency (the stress-bearing unit is necessarily the syllable)’.⁴⁴ Two additional properties are characteristic of accent, although they do not suffice to define it, as they are also found in some *bona fide* tone systems (Hyman 2006, 2010). These are CULMINATIVITY – there cannot be more than one main accent per word – and DEMARCATIVITY – accent occurs in a predictable position with reference to some morpheme edge (van der Hulst 1999, 2002, 2006; Hyman 2006; van Zanten & Goedemans 2007; Downing 2010; among others).

Only one of these properties is found in the downstep pattern of Drubea and Numèè: culminativity.⁴⁵ Indeed, there cannot be more than one downstep per stem – although a morphological word may

⁴²Rephrased from ‘Utterance-initially, there is no phonetic contrast between High tone and downstepped High tone’ to accommodate languages with downstepped M and/or L.

⁴³The exceptions include Bamileke Dschang (Hyman 1979: 12; Pulleyblank 1986: 39–42), Ikaan (Salfner 2009: 93, 96–97, 289–296), Kipare, (Odden 1986: 263–264) and Paici (Lionnet 2022b).

⁴⁴It is likely that this criterion is too strict, and that languages where the accent-bearing unit is the mora are also attested, for example, Xârâcùù (Rivierre 1978; Lionnet 2025).

⁴⁵Interestingly, culminativity of downstep is a property shared by two other Kanak languages with underlying downstep: Paici (Lionnet 2022b), a tonal language, and Xârâcùù, where downstep is accentual (Lionnet 2025; cf. Rivierre 1978). It is not the case

have more than one, as in verbs with a verbal classifier prefix (when the initial syllable of the verb is downstepped, as in (56b) and (57)), or compounds consisting of more than one stem containing a downstepped RBU (e.g., /^{*}ʊʊ+ũ+poo^{*}ka/ ‘blanket’, lit. ‘mat+hair+animal’). However, this is not a sufficient definitional criterion of accent, as we have seen. Neither of Hyman’s (2006) inviolable criteria is met: downstep is not obligatory in Drubea and Numèè, where many stems (and words) contain only registerless RBUs, that is, have no indication of register. Downstep is also not syllable-dependent: the RBU is the mora (although the syllable plays an important role in determining which mora in a CVV syllable is the RBU). Finally, downstep in Drubea and Numèè is not demarcative, since it is found on both the first and second syllables of disyllabic stems; that is, its position within a stem does not seem to be defined or constrained by a reference to either edge of the stem.

The second prototypical word-prosodic system is TONE. Hyman (2006: 229) defines a tonal language as one ‘in which an indication of pitch enters into the lexical realisation of at least some morphemes’. Crucial to this definition are the facts that the domain of tone is the morpheme, and that tones and tonal operations belong to lexical (as opposed to postlexical) phonology. Given this definition, Drubea and Numèè can be classified as *bona fide* tonal languages: their word-prosodic system is entirely based on a lexical pitch-based contrast which has a morpheme as its domain – specifically the stem (and to a certain extent the verbal classifier prefixes, which get their register specification from a lexical tonal operation). The only difference with other tone languages is that the pitch contrast is entirely syntagmatically defined in Drubea and Numèè, contrary to all other tonal languages known to date, where it is defined paradigmatically.

One must thus acknowledge that tone systems come in two types: TONE-based systems, in which tonal contrasts are defined paradigmatically, for example, H vs. L, with H having a higher pitch target than L in the same context,⁴⁶ and REGISTER-based systems, in which the tonal contrast is syntagmatically defined.⁴⁷ The separate existence of these two types, as revealed by Drubea and Numèè, justifies treating register features and tones as independent phonological objects of different types, as in the register analysis proposed in §4.

7 Conclusion

In this article, I have proposed a register-based reanalysis of the word-prosodic system of Drubea and Numèè, building on Rivierre’s (1973) initial description and analysis. This reanalysis brings to light a typologically unique tone system entirely based on register features – that is, without tones. The major contrast is indeed between downstepped and prosodically unspecified units. The behaviour of these units justifies an analysis of downstep as an underlying register feature, which is the only underlying phonological primitive needed to account for the entire word-prosodic system. I have shown that the register analysis fares better than an analysis where tone is primary and register features derived. The possibility for register features to exist in the absence of tone, including in underlying representations, is a strong argument in favour of dedicated representations for register, and of a clear separation between register features and tone in tonal representations.

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in Baga Pukur (Rochant 2023: 193), however, which seems to indicate that it might be a feature typical of New Caledonia rather than a property of underlying downstep in general.

⁴⁶It is crucial to take context into account given the relative character of the realisation of tonal contrasts: a H tone can have lower pitch in some contexts (e.g., utterance-finally) than a L tone in another context (e.g., utterance-initially), but within any given context, a H tone is realised higher than a L tone.

⁴⁷Languages with both tone and downstep and/or upstep – for example, the canonical case of downstep emerging from the syntagmatic lowering effect of a L tone on a following H tone – have elements of both types.

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Abbreviations and glosses. Abbreviations in this article follow the Leipzig Glossing Rules, with the following exceptions:

ACT	active
ASSERT	assertive
COLL	collective
DESCR	descriptive
EPIST	epistemic
PROS	prospective
RBU	register-bearing unit
STAT	stative
TBU	tone-bearing unit
TRN	tonal root note

References

- Académie des Langues Kanak (2011). *Proposition d'écriture du nââ kwènyî*. Nouméa: Académie des Langues Kanak.
- Académie des Langues Kanak (2015). *Proposition d'écriture du nââ numèè*. Nouméa: Académie des Langues Kanak.
- Académie des Langues Kanak (2018). *Proposition d'écriture du naa drubea*. Nouméa: Académie des Langues Kanak.
- Académie des Langues Kanak (2023). Apprendre le drubea. Online language tutorial published at <https://www.drubea.com>.
- Adeniyi, Kolawole (2016). Downstep in Igala and Yala (Ikoma). *Journal of West African Languages* 43, 1–21.
- Andersen, Torben (1988). Downstep in Pàri: the tone system of a Western Nilotic language. *Studies in African Linguistics* 19, 261–315.
- Anderson, Stephen C. & Jeanette Swackhamer (1981). From consonants to downstep in Podoko. *Studies in African Linguistics* 12, 131–153.
- Ati, Marguerite & Jean-Claude Rivierre (1966). *Le fils de l'aigle pêcheur*. Numèè corpus, Pangloss Collection, LACITO-CNRS.
- Beavon-Ham, Virginia (2012). Consonant–tone interaction in Saxwe. In Michael R. Marlo (ed.) *Selected proceedings of the 42nd Annual Conference on African Linguistics*. Somerville, MA: Cascadia, 55–69.
- Bird, Steven & Oliver Stegen (1993). Tone in the Bamileke Dschang associative construction: an electrolaryngographic study and comparison with Hyman (1985). Technical Report RP-57, Centre for Cognitive Science, University of Edinburgh.
- Bird, Steven & Oliver Stegen (1995). The Bamileke Dschang associative construction: instrumental findings. Technical Report RP-66, Centre for Cognitive Science, University of Edinburgh.
- Boyeldieu, Pascal (2000). *La langue bagiro (République Centrafricaine)*. Frankfurt: Peter Lang.
- Boyeldieu, Pascal (2019). Proper names and case markers in Sinyar (Chad/Sudan). *STUF – Language Typology and Universals* 72, 467–504.
- Carlson, Robert (1983). Downstep in Supyire. *Studies in African Linguistics* 14, 35–45.
- Clements, G.N. (1979). The description of terraced-level tone languages. *Lg* 55, 536–558.
- Clements, G.N. (1981). The hierarchical representation of tone features. In G.N. Clements (ed.) *Harvard studies in phonology*, volume 2. Bloomington, IN: Indiana University Linguistics Club, 187–272.
- Clements, G.N. & Kevin C. Ford (1977). On the phonological status of downstep in Kikuyu. In G.N. Clements (ed.) *Harvard studies in phonology*, volume 1. Cambridge, MA: Department of Linguistics, Harvard University, 187–272.
- Clements, G.N. & Elizabeth V. Hume (1995). The internal organization of speech sounds. In John A. Goldsmith (ed.) *The handbook of phonological theory*. Oxford: Blackwell, 245–306.
- Connell, Bruce (2011). Downstep. In Marc van Oostendorp, Colin J. Ewen, Keren Rice & Elizabeth V. Hume (eds.) *The Blackwell companion to phonology*. Oxford: Wiley-Blackwell, 824–847.
- Creider, Chet A. (1982). *Studies in Kalenjin nominal tonology*. Berlin: Reimer.
- Daly, John P. & Larry M. Hyman. (2007). On the representation of tones in Peñoles Mixtec. *IJAL* 73, 165–208.
- Downing, Laura J. (2010). Accent in African languages. In Harry van der Hulst, Rob Goedemans & Ellen van Zanten (eds.) *A survey of word accentual patterns in the languages of the world*. Berlin: Mouton de Gruyter, 381–427.
- Elugbe, Ben Ohi (1986). The analysis of falling tones in Ghotuo. In Koen Bogerts, Harry van der Hulst & Maarten Mous (eds.) *The phonological representation of suprasegmentals*. Dordrecht: Foris, 51–62.
- Genet, Marcelle, Yvonne Païta, Soeur Caroline, Marguerite Wamytan, Augustine Betto & Tadahiko L.A. Shintani (1992). *Textes en nrâa drubea*. Tokyo: Institut de Recherches sur les Langues et Civilisations d'Asie et d'Afrique (ILCAA).
- Gordon, Matthew & Ian Maddieson (1999). The phonetics of Ndumbea. *Oceanic Linguistics* 38, 66–90.
- Han, Miek S. & Kong-On Kim (1974). Phonetic variation of Vietnamese tones in disyllabic utterances. *JPh* 2, 223–232.
- Haudricourt, André-Georges (1971). New Caledonia and the Loyalty Islands. In J. Donald Bowen (ed.) *Linguistics in Oceania*, volume 1. Berlin: Mouton de Gruyter, 359–396.
- Haudricourt, André-Georges & Jean-Claude Rivierre (n.d.). Numèè: récits recueillis par André-Georges Haudricourt et Jean-Claude Rivierre. Pangloss Collection (CNRS). Published online at <https://pangloss.cnrs.fr/corpus/Num>

- van der Hulst, Harry (1999). Word accent. In Harry van der Hulst (ed.) *Word prosodic systems in the languages of Europe*. Berlin: Mouton de Gruyter, 3–115.
- van der Hulst, Harry (2002). Stress. In Lynn Nadel (ed.) *Encyclopedia of cognitive science*, volume 4. London: Nature Publishing Group, 246–254.
- van der Hulst, Harry (2006). Word stress. In Keith Brown (ed.) *Encyclopedia of language and linguistics*, 2nd edition. Amsterdam: Elsevier, 655–665.
- Hyman, Larry M. (1979). A reanalysis of tonal downstep. *Journal of African Languages and Linguistics* 1, 9–29.
- Hyman, Larry M. (1984). Book review: *Studies in Kalenjin Nominal Tonology* by Chet A. Creider. *Lg* 60, 674–675.
- Hyman, Larry M. (1985a). *A theory of phonological weight*. Dordrecht: Foris.
- Hyman, Larry M. (1985b). Word domains and downstep in Bamileke-Dschang. *Phonology Yearbook* 2, 45–82.
- Hyman, Larry M. (1986). The representation of multiple tone heights. In Koen Bogers, Harry van der Hulst & Maarten Mous (eds.) *The phonological representation of suprasegmentals: studies on African languages offered to John M. Stewart on his 60th birthday*. Dordrecht: Foris, 109–152.
- Hyman, Larry M. (1993). Register tones and tonal geometry. In Keith Snider & Harry van der Hulst (eds.) *The phonology of tone: the representation of tonal register*. Berlin: Mouton de Gruyter, 75–108.
- Hyman, Larry M. (2006). Word-prosodic typology. *Phonology* 23, 225–257.
- Hyman, Larry M. (2010). Do tones have features? In John A. Goldsmith, Elizabeth Hume & W. Leo Wetzels (eds.) *Tones and features: phonetic and phonological perspectives*. Berlin: Mouton de Gruyter, 50–80.
- Hyman, Larry M. (2017). Synchronic vs. diachronic naturalness: Hyman and Schuh (1974) revisited. *UC Berkeley Phonetics and Phonology Lab Annual Report* 13, 226–242.
- Hyman, Larry M. & Russell Schuh (1974). Universals of tone rules: evidence from West Africa. *LI* 5, 81–115.
- Inkelas, Sharon (1989). Register tone and the phonological representation of downstep. In Isabelle Haik & Laurice Tuller (eds.) *Current approaches to African linguistics*. Berlin: De Gruyter Mouton, 65–82.
- Inkelas, Sharon, William R. Leben & Mark Cobler (1987). The phonology of intonation in Hausa. *NELS* 17, 327–341.
- INSEE-ISEE (2019). Recensement de la population de Nouvelle-Calédonie. Census, Institut National de la Statistique et des Etudes Economiques (Paris, France), Institut de la Statistique et des Etudes Economiques (Nouméa, Nouvelle-Calédonie).
- Kim, Eui-Jung (1996). *Edopi phonology*. Ms., Summer Institute of Linguistics.
- Leben, William R. (2018). *The nature(s) of downstep*. Ms., Stanford University.
- Leenhardt, Maurice (1946). *Langues et dialectes de l'Austro-Mélanésie*. Paris: Institut d'Ethnologie.
- Lightfoot, Nancy (1974). Tones on Kru monosyllables. *Anthropological Linguistics* 16, 425–441.
- Lionnet, Florian (2022a). The features and geometry of tone in Laal. *Phonology* 39, 251–292.
- Lionnet, Florian (2022b). Tone and downstep in Paicî (Oceanic, New Caledonia). *Phonological Data and Analysis* 4, 1–47.
- Lionnet, Florian (2025). Accent and tone: the double origin of the Paicî tone system. *Diachronica* 42, 406–427.
- Litvinova, Lora (2023). *A grammatical analysis and documentation of Kugama, an Adamawa language of Nigeria*. PhD dissertation, Institut national des langues et civilisations orientales.
- Lynch, David, Malcolm Ross & Terry Crowley (2002). Internal subgrouping. In David Lynch, Malcolm Ross & Terry Crowley (eds.) *The Oceanic languages*. Richmond, Surrey: Curzon, 92–120.
- Miller, Amanda L. (2010). A prosodic account of Ju'hoansi consonant distributional asymmetries. In Matthias Brenzinger & Chirsta König (eds.) *Khoisan languages and linguistics: proceedings of the 1st international symposium*. Cologne: Rüdiger Köppe, 53–84.
- Nicole, Jacques (1980). Downstepped low tone in Nawdm. *Journal of African Languages and Linguistics* 2, 133–139.
- Nissim, Gabriel M. (1981). *Le bamileke-ghomálá' (parler de Bandjoun, Cameroun)*. Paris: Centre National de la Recherche Scientifique.
- Odden, David (1982). Tonal phenomena in Kishambaa. *Studies in African Linguistics* 13, 177–208.
- Odden, David (1986). On the Obligatory Contour Principle. *Lg* 62, 353–383.
- Pulleyblank, Douglas (1986). *Tone in lexical phonology*. Dordrecht: D. Reidel.
- Rendina, Sophie (2009). *Description of Numee*. Master's thesis, SOAS University of London.
- Rialland, Annie (1997). Le parcours du “downstep” ou l'évolution d'une notion. *Bulletin de la Société de Linguistique de Paris* 92, 207–243.
- Rialland, Annie (2001). Anticipatory raising in downstep realization: evidence for preplanning in tone production. In Shigeki Kaji (ed.) *Proceedings of the symposium cross-linguistic studies of tonal phenomena: tonogenesis, Japanese accentology, and other topics*. Tokyo: Institute for the Study of Languages and Cultures of Asia and Africa (ILCAA), 301–322.
- Rivierre, Jean-Claude (1973). *Phonologie comparée des dialectes de l'extrême-sud de la Nouvelle Calédonie*. Paris: SELAF.
- Rivierre, Jean-Claude (1974). Tons et segments du discours en paicî (Nouvelle-Calédonie). *Bulletin de la Société de Linguistique de Paris* 69, 325–340.
- Rivierre, Jean-Claude (1978). Accents, tons et inversion tonale en Nouvelle-Calédonie. *Bulletin de la Société de Linguistique de Paris* 73, 415–443.
- Rivierre, Jean-Claude (1980). *La langue de Touho: phonologie et grammaire du cèmuhi (Nouvelle Calédonie)*. Paris: Peeters.
- Rivierre, Jean-Claude & Clément Vandégou (n.d.). *Lexique numèè-français*. Ms., LACITO-CNRS.
- Rochant, Neige (2023). *Bilectal grammar of Baga Pukur, an Atlantic language of Guinea*. PhD dissertation, Université Sorbonne Nouvelle.

- Rolle, Nicholas (2018). *Grammatical tone: typology and theory*. PhD dissertation, University of California, Berkeley.
- Sagey, Elizabeth (1986). *The representation of features and relations in nonlinear phonology*. PhD dissertation, Massachusetts Institute of Technology.
- Salfner, Sophie (2009). *Tone in the phonology, lexicon and grammar of Ikaan*. PhD dissertation, SOAS University of London.
- Shintani, Tadahiko L.A. (2019). *Analyse du parler nráa drübea*. Noumea: Académie des Langues Kanak.
- Shintani, Tadahiko L. A. & Yvonne Païta (1990a). *Dictionnaire de la langue de païta*. Nouméa: Société d'études historiques de Nouvelle-Calédonie.
- Shintani, Tadahiko L.A. & Yvonne Païta (1990b). *Grammaire de la langue de païta*. Noumea: Société d'études historiques de Nouvelle-Calédonie.
- Snider, Keith L. (1990). Tonal upstep in Krachi: evidence for a register tier. *Lg* **66**, 453–474.
- Snider, Keith L. (1999). *The geometry and features of tone*. 1st edition. Dallas, TX: SIL International and University of Texas at Arlington.
- Snider, Keith L. (2020). *The geometry and features of tone*. 2nd edition. Dallas, TX: SIL International.
- Soon, Samantha Sze Min (2023). *Kwényi: a sketch grammar from a historical perspective*. PhD dissertation, University of Sydney.
- Stewart, John M. (1965). The typology of the Twi tone system. *Bulletin of the Institute of African Studies* **1**, 1–27.
- Stewart, John M. (1983). Downstep and floating low tones in Adioukrou. *Journal of African Languages and Linguistics* **5**, 57–78.
- Stewart, John M. (1993). Dschang and Ebrie as Akan-type total downstep languages. In Keith Snider & Harry van der Hulst (eds.) *The phonology of tone: the representation of tonal register*. Berlin: Mouton de Gruyter, 185–244.
- Tadadjeu, Maurice (1974). Floating tones, shifting rules, and downstep in Dschang-Bamileke. *Studies in African Linguistics* **5**, 283–290.
- Wacalie, Fabrice (2013). *Description morpho-syntaxique du nâa numèè (langue de Yaté, Extrême-Sud de la Nouvelle-Calédonie)*. PhD dissertation, Institut national des langues et civilisations orientales.
- Wang, William S.-Y. (1967). Phonological features of tone. *IJAL* **33**, 93–105.
- Winston, F.D.D. (1960). The 'mid tone' in Efik. *African Language Studies* **1**, 185–192.
- Xu, Yi (1997). Contextual tonal variations in Mandarin. *JPh* **25**, 61–83.
- Yip, Moira (1980). *The tonal phonology of Chinese*. PhD dissertation, Massachusetts Institute of Technology.
- Yip, Moira (1989). Contour tones. *Phonology* **6**, 149–174.
- van Zanten, Ellen & Rob Goedemans (2007). A functional typology of Austronesian and Papuan stress systems. In Vincent J. van Heuven & Ellen van Zanten (eds.) *Prosody in Indonesian languages*. Utrecht: LOT, 63–88.